



Technical University of Applied Sciences
Würzburg-Schweinfurt

Basics of Electronics and Components

1. Scalar and Vector Fields

Prof. Dr. Henning Lormann

office 1.1.20

phone +49 9721 940-8347

mail henning.lormann@thws.de

© Prof. Dr. Henning Lormann, 2026

All rights reserved. No part of this material may be reproduced
or distributed without the prior permission of the author.

Learning objectives

After this chapter, you will be able to:

- Distinguish between scalars and vectors and classify examples from electrical engineering
- Apply basic vector operations (addition, scalar multiplication, scalar product, cross product) correctly
- Interpret scalar fields and vector fields using arrow plots and field lines
- Calculate gradient, divergence, and curl for given fields
- Explain the physical meaning of grad, div, and rot in electromagnetic contexts

1. Scalar and vector fields

Chapter outline

Chapter outline

- Scalars and Vectors: Definitions and examples
- Coordinate Systems: Cartesian, cylindrical, spherical
- Vector Algebra: Basic operations

1.1 Fundamentals of linear algebra

Basics of electronics and components
© Prof. Dr. Henning Lormann, 2026

5

Chapter outline

- Scalar fields and vector fields
- Vector Calculus: ∇ -operator, grad, div, rot
- Applications to electromagnetic fields

1.2 Vector calculus

Basics of electronics and components
© Prof. Dr. Henning Lormann, 2026

22

1. Scalar and vector fields

1. Scalar and vector fields

Lower case greek letters & commonly used capitals

α	alpha	ι	iota	ρ	rho
β	beta	κ	kappa	$\sigma \Sigma$	sigma
$\gamma \Gamma$	gamma	$\lambda \Lambda$	lambda	τ	tau
$\delta \Delta$	delta	μ	mu	$\upsilon \Upsilon$	upsilon
$\epsilon (\varepsilon)$	epsilon ('varepsilon')	ν	nu	$\phi (\varphi) \Phi$	phi ('varphi')
ζ	zeta	$\xi \Xi$	xi	χ	chi
η	eta	$\omicron$	omicron	$\psi \Psi$	psi
$\theta \Theta$	theta	$\pi \Pi$	pi	$\omega \Omega$	omega

Chapter outline

- Scalars and Vectors: Definitions and examples
- Coordinate Systems: Cartesian, cylindrical, spherical
- Vector Algebra: Basic operations

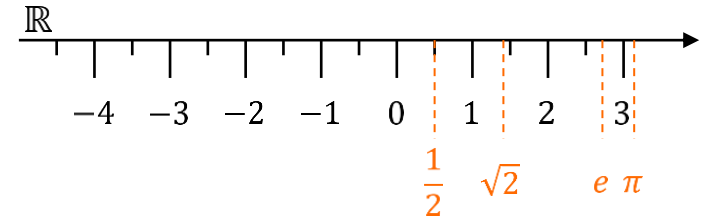
1.1 Fundamentals of linear algebra

1.1 Fundamentals of linear algebra

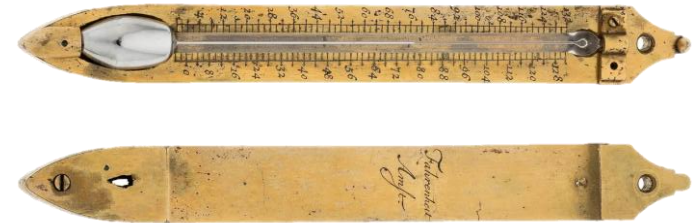
Scalars

- A scalar is a **single real number**, represented as a point on the **one-dimensional real number line** $\mathbb{R}$.
- It has **one component** and can be described by its **magnitude** (also length, amplitude, or absolute value).
- In a **vector space**, a scalar **scales a vector**:
 - ⌋ It produces another vector with **scaled length**.
 - ⌋ It **does not change the direction** of the vector, (when the scalar is negative, it scales in the opposite direction).
- Operation rules
 - ⌋ Commutative: $a + b = b + a$
 - ⌋ Associative: $a + (b + c) = (a + b) + c$
 - ⌋ Distributive: $x \cdot (y + z) = x \cdot y + x \cdot z$
- A **scalar quantity** describes a physical property by its magnitude and unit and has no direction.
- **Physical examples**: Temperature, electric potential, pressure

? Do you know other scalar quantities and can you describe their meaning.
Quick check



Representation of different scalar values on the real number line $\mathbb{R}$.



*Example of temperature as scalar value :
Original mercury thermometer made in 1714
by Daniel Fahrenheit.*

1.1 Fundamentals of linear algebra

Cartesian coordinate system

- 3D space requires **three directions** defined by **three unit vectors** (usually right-hand-systems).

- In a **Cartesian coordinate system** x, y and z are defined by three **orthogonal unit vectors**:

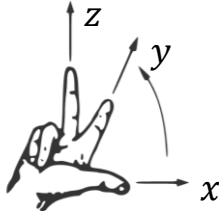
$$\vec{e}_x = (1, 0, 0), \quad \vec{e}_y = (0, 1, 0) \quad \text{and} \quad \vec{e}_z = (0, 0, 1)$$

- The **cross product** of two unit vectors generate the **third basis vector**.

$$\vec{e}_x \times \vec{e}_y = \vec{e}_z, \quad \vec{e}_y \times \vec{e}_z = \vec{e}_x, \quad \text{and} \quad \vec{e}_z \times \vec{e}_x = \vec{e}_y$$

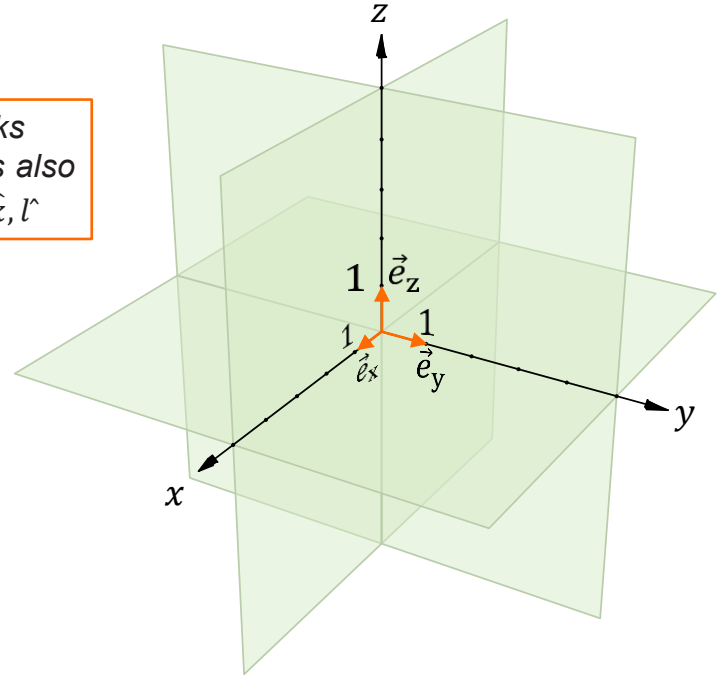
- Use your right hand to construct a **right-hand coordinate system**:

- thumb = x ,
- index = y
- middle = z



- Note: Besides Cartesian coordinates, cylindrical and spherical systems are also widely used.*

*In textbooks
sometimes also
 $\hat{j}, \vec{k}, \vec{l}$ or $\hat{j}, \hat{k}, \hat{l}$*



A Cartesian coordinate system defined by the unit vectors $\vec{e}_x$, $\vec{e}_y$ and $\vec{e}_z$.

1.1 Fundamentals of linear algebra

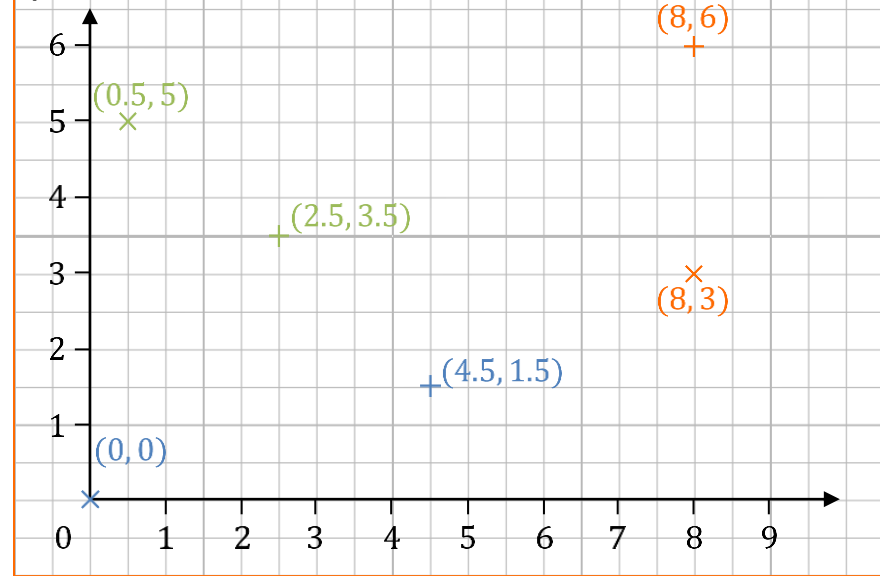
Vector basics

- A vector $\vec{r}$ is an **element of an n-dimensional real space $\mathbb{R}^n$** .
 - ⌈ $\mathbb{R}^1$: one dimension
 - ⌈ $\mathbb{R}^2$: two dimensions (can be identified with complex numbers: $(x, y) \leftrightarrow x + jy$)
 - ⌈ $\mathbb{R}^3$: three dimensions. Describes our physical reality.
- In $\mathbb{R}^2$ and $\mathbb{R}^3$, **represented as arrows**.
- A vector has:
 - ⌈ **Coordinates**: the scalar components $(r_1, r_2, \dots, r_n)$ in a coordinate system
 - ⌈ **Direction**: the unit vector $\frac{\vec{r}}{\|\vec{r}\|}$ (for $\vec{r} \neq 0$)
 - ⌈ **Norm $\|\vec{r}\|$** : the length of the vector (a non-negative real number)
- **Vector quantities** describe physical phenomena by their magnitude and direction.
- **Physical examples**: electric field, magnetic field

A scalar can be seen as a '1D vector'.

?
Quick check
1. Name three different vector quantities.
2. What makes them different from scalar quantities?

Example: Identify the vectors starting at $\times$ and ending at $+$ and their norm between the marked points.



A set of vectors $\vec{a}$, $\vec{b}$ and $\vec{c}$ in $\mathbb{R}^2$.

1.1 Fundamentals of linear algebra

Position vector

- A **position vector** $\vec{r}$ points **from the origin** $(0, 0, 0)$ **to a point** in space (x, y, z)
- In the **Cartesian basis** $(\vec{e}_x, \vec{e}_y, \vec{e}_z)$:

$$\vec{r} = \begin{bmatrix} x \\ y \\ z \end{bmatrix} = (x, y, z) = x \vec{e}_x + y \vec{e}_y + z \vec{e}_z$$

Matrix notation

Tuple notation

Unit vector notation

- Euclidean **norm** (i. e. its length) $\|\vec{r}\|$ is 'Pythagorean theorem in 3D'.

$$\|\vec{r}\| = \sqrt{x^2 + y^2 + z^2}$$

- **Note:** If clear from context, set $r := \|\vec{r}\|$. Avoid $|\vec{r}|$ (single bars are for scalars, e. g. if $x = -1$, then $|x| = 1$).

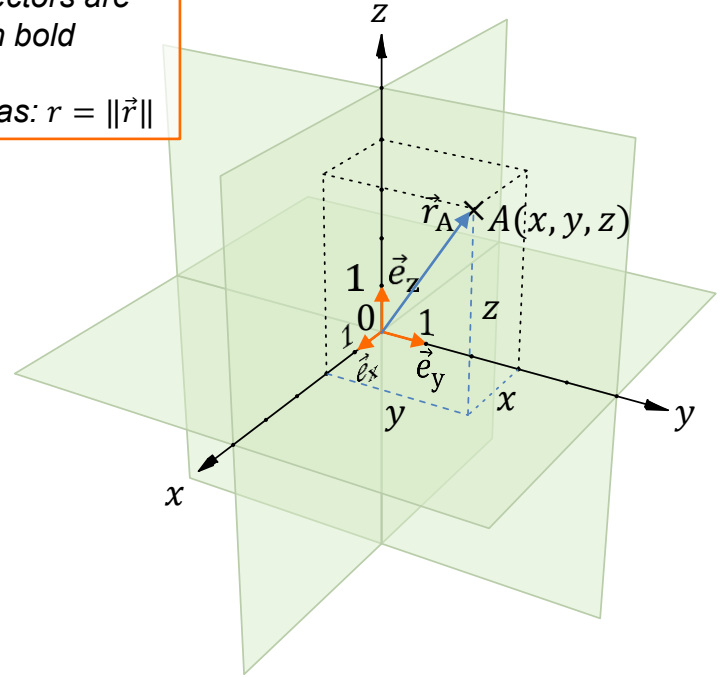


$|x|$ (and $|z|$ for complex z) gives the distance to zero on the number line/complex plane; a norm $\|\vec{r}\|$ does the same for vectors, i. e. it assigns a non-negative length to each vector in a consistent way.



In textbook, vectors are often written in bold letters: $\mathbf{r} = \vec{r}$

Vector norms as: $r = \|\vec{r}\|$



Vector $\vec{r}_A$ pointing from the origin $(0, 0, 0)$ to point $A(x, y, z)$.

1.1 Fundamentals of linear algebra

Displacement vector

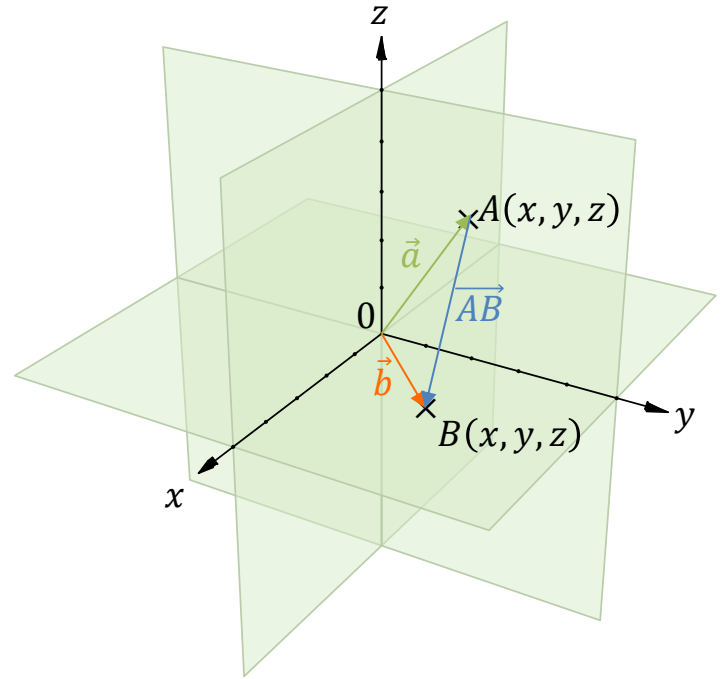
- A **displacement vector** $\overrightarrow{AB}$ connects two points A and B .
- To compute the displacement vector from point A to point B from the position vectors $\vec{a}$ and $\vec{b}$ by subtracting **start from end**

$$\overrightarrow{AB} = \vec{b} - \vec{a} = \begin{bmatrix} x_B - x_A \\ y_B - y_A \\ z_B - z_A \end{bmatrix}$$

- The distance formula $\|\overrightarrow{AB}\|$ applies.

$$\|\overrightarrow{AB}\| = \sqrt{(x_B - x_A)^2 + (y_B - y_A)^2 + (z_B - z_A)^2}$$

- *Example: Calculate the vector $\overrightarrow{AB}$ pointing from $A(2, 2, 4)$ to $B(2, 2, 0)$. Then calculate the norm $\|\overrightarrow{AB}\|$ of the vector*



Three-dimensional representation of vector $\overrightarrow{AB}$ in a Cartesian coordinate system.

1.1 Fundamentals of linear algebra

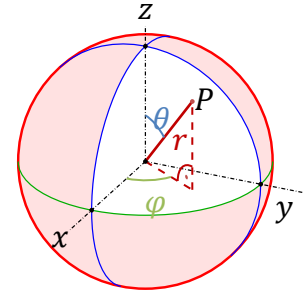
Spherical and cylindrical coordinate systems

- For some practical problems **switching from a Cartesian** to spherical or cylindrical coordinates comes handy.
- Spherical coordinates** (radial symmetry around point charges)
 - Basis vectors: $\vec{e}_r$ (radius), $\vec{e}_\varphi$ (azimuth), $\vec{e}_\theta$ (polar angle)
 - Position vector and length: $\vec{r} = r \vec{e}_r$ and $r := \|\vec{r}\|$
 - Relation to Cartesian coordinates

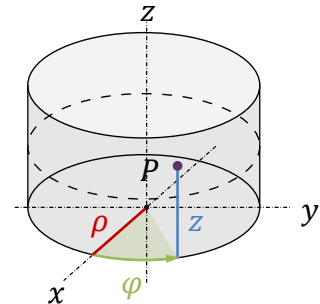
$$\begin{aligned} x &= r \cos \varphi \sin \theta, & \text{with } 0 \leq r < \infty \\ y &= r \sin \varphi \sin \theta & 0 \leq \varphi \leq 2\pi \\ z &= r \cos \theta & 0 \leq \theta \leq \pi \end{aligned}$$

- Cylindrical coordinates** (axial symmetry around wires)
 - Basis vectors: $\vec{e}_\rho$ (radius), $\vec{e}_\varphi$ (angular coord.), $\vec{e}_z$ (height)
 - Position vector: $\vec{r} = \rho \vec{e}_\rho + z \vec{e}_z$ and $\|\vec{r}\| = \sqrt{\rho^2 + z^2}$
 - Relation to Cartesian coordinates

$$\begin{aligned} x &= \rho \cos \varphi, & \text{with } 0 \leq \rho < \infty \\ y &= \rho \sin \varphi & 0 \leq \varphi \leq 2\pi \\ z &= z \end{aligned}$$



Spherical coordinates (r, θ, φ) as commonly used around point charges.



Cylindrical coordinates (ρ, φ, z) as commonly used around wires.

1.1 Fundamentals of linear algebra

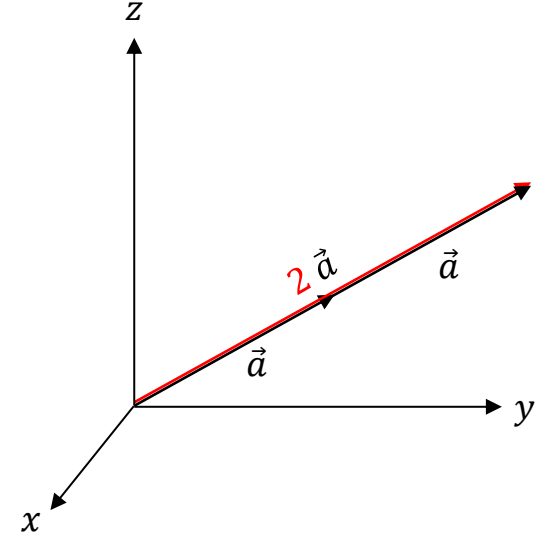
Scalar multiplication

- A scalar λ **'scales'** a vector $\vec{a}$ without changing its direction

- Associative: $\lambda \vec{a} = \vec{a} \lambda = \lambda \begin{bmatrix} a_x \\ a_y \\ a_z \end{bmatrix} = \begin{bmatrix} \lambda a_x \\ \lambda a_y \\ \lambda a_z \end{bmatrix}$

- Distributive: $(\lambda + \gamma)\vec{a} = \lambda \vec{a} + \gamma \vec{a}$

- *Example: Multiply the vector $\vec{a} = (1, 2, 3)$ by 0, 1, -1 , 0.5 and 2. What properties do the resulting vectors have.*



Three-dimensional representation of the scalar multiplication of vector $\vec{a}$ with $\lambda = 2$.

1.1 Fundamentals of linear algebra

Vector addition

- Two vectors are added piece by piece

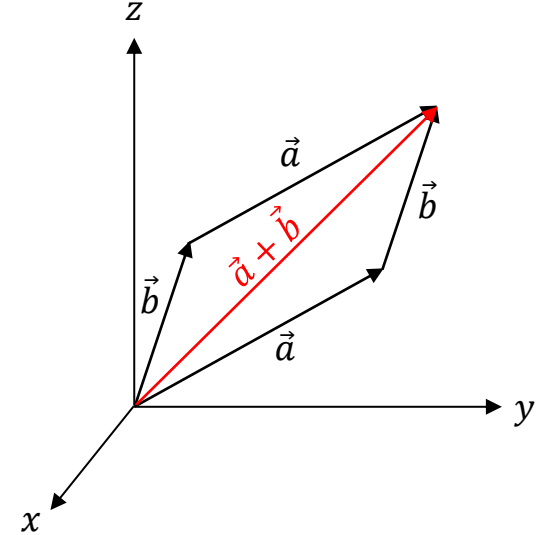
$$\vec{a} + \vec{b} = \begin{bmatrix} a_x \\ a_y \\ a_z \end{bmatrix} + \begin{bmatrix} b_x \\ b_y \\ b_z \end{bmatrix} = \begin{bmatrix} a_x + b_x \\ a_y + b_y \\ a_z + b_z \end{bmatrix}$$

- Properties

- Associative: $\vec{a} + \vec{b} = \vec{b} + \vec{a}$

- Distributive: $\lambda (\vec{a} + \vec{b}) = \lambda \vec{a} + \lambda \vec{b}$

- Example: Add the two vectors $\vec{a} = (1, 2, 3)$ and $\vec{b} = (4, 5, 6)$.
Add the two vectors $\vec{a}$ and $-\vec{a}$. Add the zero vector $\vec{0}$ to $\vec{a}$.



Three-dimensional representation of the addition of two vectors $\vec{a}$ and $\vec{b}$.

1.1 Fundamentals of linear algebra

Scalar product



Do **not** confuse **scalar multiplication** $\lambda \vec{a}$ with the **scalar product**. The former yields another vector, the latter a scalar!

- Also called **scalar product** (also dot or inner product) $\vec{a} \cdot \vec{b}$ measures **how much $\vec{a}$ points in the direction of $\vec{b}$** .
- In Cartesian components with the angle ϕ between $\vec{a}$ and $\vec{b}$

$$\vec{a} \cdot \vec{b} = a_x \cdot b_x + a_y \cdot b_y + a_z \cdot b_z = \|\vec{a}\| \|\vec{b}\| \cos \phi$$

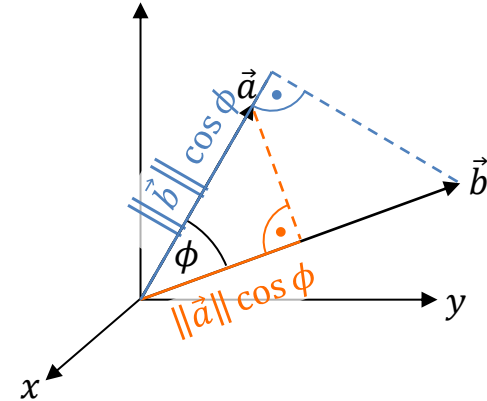
Intuition: $\vec{a}$ casts a 'shadow' on $\vec{b}$ (or vice versa); the shadow length (scalar projection) is $\|\vec{a}\| \cos \phi$

- Special cases

$$\vec{a} \cdot \vec{b} = 0 \quad \text{for } \phi = 90^\circ \text{ (orthogonal } \perp)$$

$$\vec{a} \cdot \vec{b} = \|\vec{a}\| \|\vec{b}\| \quad \text{for } \phi = 0^\circ \text{ (codirectional } \parallel)$$

- Note: The scalar product yields a scalar value **not a vector**!
- Example: Calculate $\|\vec{a}\| \cdot \|\vec{b}\|$ and the scalar product of $\vec{a} = (0, 1, 1)$, $\vec{b} = (1, 0, 2)$ and $\vec{c} = (2, 1, -1)$. What do you notice.



Dot product of the two vectors $\vec{a}$ and $\vec{b}$.

1.1 Fundamentals of linear algebra

Cross product

- A cross product (also vector product) produces a **new vector perpendicular** (normal vector $\vec{n}$) to the plane containing $\vec{a}$ and $\vec{b}$:

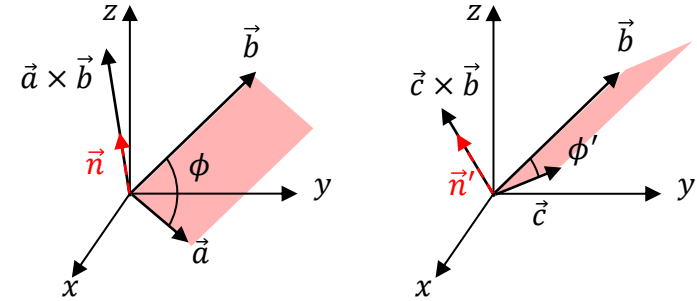
$$\vec{a} \times \vec{b} = \begin{bmatrix} a_x \\ a_y \\ a_z \end{bmatrix} \times \begin{bmatrix} b_x \\ b_y \\ b_z \end{bmatrix} = \begin{bmatrix} a_y b_z - a_z b_y \\ a_z b_x - a_x b_z \\ a_x b_y - a_y b_x \end{bmatrix} = \|\vec{a}\| \|\vec{b}\| \sin \phi \vec{n}$$

- The **length** of $\vec{a} \times \vec{b}$ is **proportional to the area** spanned by the two vectors. The normal $\vec{n}$ to the surface is orthogonal to both $\vec{a}$ and $\vec{b}$.

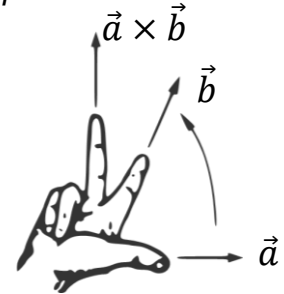
$$\vec{a} \times \vec{b} = \|\vec{a}\| \|\vec{b}\| \vec{n} \quad \text{for } \phi = 90^\circ \text{ (orthogonal)}$$

$$\vec{a} \times \vec{b} = 0 \quad \text{for } \phi = 0^\circ \text{ (codirectional)}$$

- Anticommutative: $\vec{a} \times \vec{b} = -\vec{b} \times \vec{a}$
- Distributive: $\vec{a} \times (\vec{b} + \vec{c}) = \vec{a} \times \vec{b} + \vec{a} \times \vec{c}$
- Example: Vector product of $\vec{a} = (1, 2, 0)$ and $\vec{b} = (4, 5, 0)$.



Cross product of two vectors producing a vector normal to the plane.

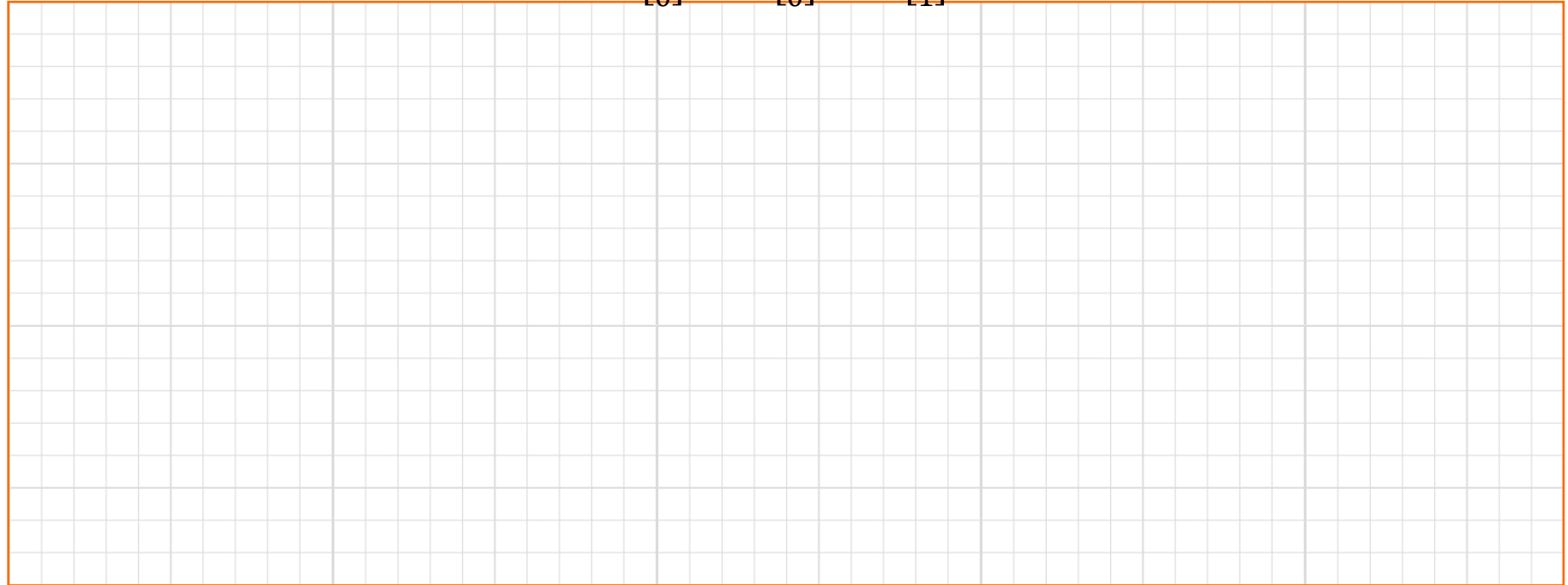


Right-hand rule to generate the direction of a vector product.

1.1 Fundamentals of linear algebra

Homework: Vector products 1/2

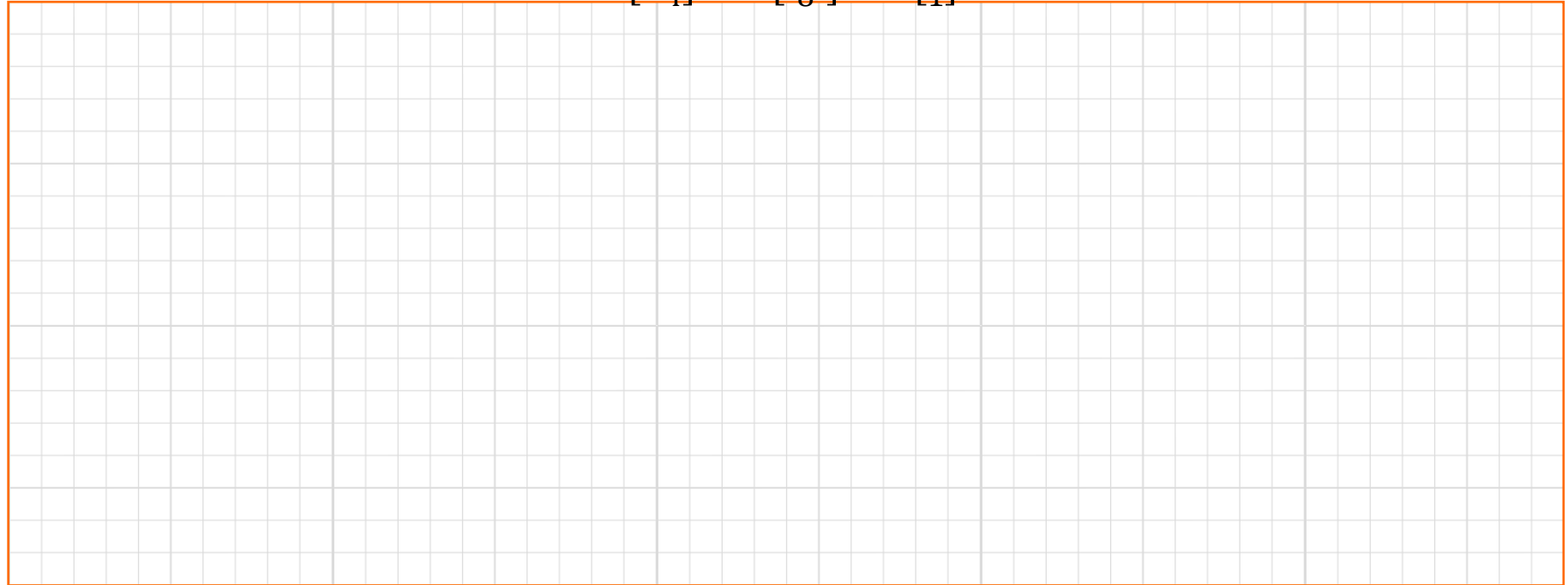
Calculate the dot and cross products of $\vec{e}_x = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$, $\vec{e}_y = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}$, $\vec{e}_z = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$. Discuss the result.



1.1 Fundamentals of linear algebra

Homework: Vector products 2/2

Calculate the dot and cross products of $\vec{a} = \begin{bmatrix} 2 \\ 3 \\ -4 \end{bmatrix}$, $\vec{b} = \begin{bmatrix} -4 \\ -6 \\ 8 \end{bmatrix}$, $\vec{c} = \begin{bmatrix} 2 \\ 0 \\ 1 \end{bmatrix}$. Discuss the result.



1.1 Fundamentals of linear algebra

Working with infinitesimally small elements

- Analysing physical problems, sometimes **infinitesimally small elements** are utilized.

- An infinitesimal **length element** $d\vec{r}$ is a vector

$$d\vec{r} = dx \vec{e}_x + dy \vec{e}_y + dz \vec{e}_z$$

Whatever symbol (describing a quantity) is behind 'd' is '**next-to-zero**' small. This will be important for integration and differentiation.

- An infinitesimal **area element** $d\vec{A}$ is defined by the area and the normal to this area:

$$d\vec{A}_x = \vec{e}_x dy dz$$

$$d\vec{A}_y = \vec{e}_y dx dz$$

$$d\vec{A}_z = \vec{e}_z dx dy$$



The cross product of two unit vectors is a vector perpendicular to both, with magnitude equal to the area they span.

- An infinitesimal **volume element** dV is a scalar:

$$dV = dx dy dz$$



The dot product of the area normal with the third unit vector gives a scalar (i. e. the volume).

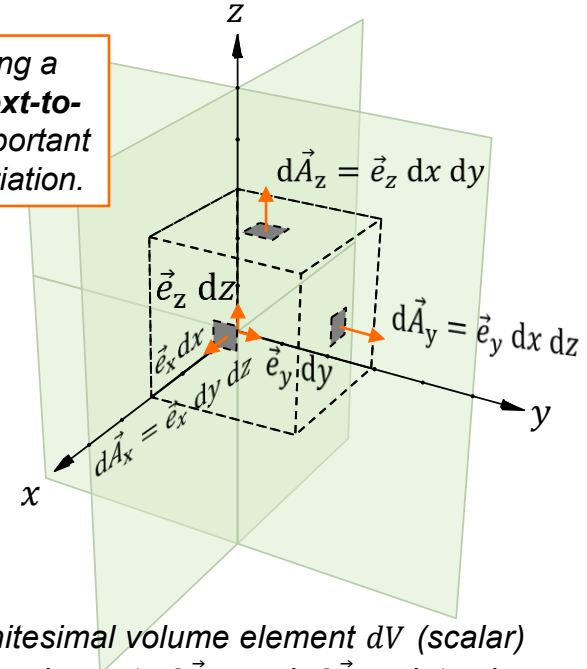


Illustration of an infinitesimal volume element dV (scalar) and the oriented area elements $d\vec{A}_i$; each $d\vec{A}_i$ points along the outward normal and is orthogonal to the two edge vectors spanning its face.

1.1 Fundamentals of linear algebra

Vector algebra

Operation	Notation	Description
Vector addition	$\vec{a} + \vec{b}$	Addition of two vectors, yielding a vector, $\mathbb{R}^n \rightarrow \mathbb{R}^n$
Scalar multiplication	$\lambda \vec{a}$	Multiplication of a scalar and a vector, yielding a vector, $\mathbb{R}^n \rightarrow \mathbb{R}^n$
Scalar product	$\vec{a} \cdot \vec{b}$	Multiplication of two vectors, yielding a scalar, $\mathbb{R}^n \rightarrow \mathbb{R}$
Vector product	$\vec{a} \times \vec{b}$	Multiplication of two vectors, yielding a vector, $\mathbb{R}^n \rightarrow \mathbb{R}^n$

Summary

- Calculus is performed by using real numbers.
- **Real numbers** $\mathbb{R}$ contain all numbers except for complex numbers.
- A scalar gives the magnitude or scalar value of a quantity. One dimension, only.
- A vector describes the Euclidean length and direction of a quantity. One or more dimensions.
- Physical in space can be calculated in Cartesian, spherical or cylindrical coordinate systems.
- In Cartesian coordinates, the length of a vector $\vec{r}$ is defined by the distance formula $\|\vec{r}\| = \sqrt{x^2 + y^2 + z^2}$
- The scalar product between two vectors yields a scalar. $\vec{a} \cdot \vec{b} = 0$, for $\vec{a} \perp \vec{b}$ and $\vec{a} \cdot \vec{b} = \|\vec{a}\| \|\vec{b}\|$, for $\vec{a} \parallel \vec{b}$
- The cross product between two vectors yields a new vector orthogonal to the two vectors. $\vec{a} \times \vec{b} = 0$, for $\vec{a} \parallel \vec{b}$ / $\vec{a} \times \vec{b} = \|\vec{a}\| \|\vec{b}\| \vec{n}$, for $\vec{a} \perp \vec{b}$

1.1 Fundamentals of linear algebra

Chapter outline

- Scalar fields and vector fields
- Vector Calculus: ∇ -operator, grad, div, rot
- Applications to electromagnetic fields

1.2 Vector calculus

1.2.1 Scalar and vector fields

1.2.1 Scalar and vector fields

Motivation: Why Fields?

— So far:

- Individual scalars (numerical value) and vectors (arrow)
- Scalar quantities (e. g. Temperature T) and vector quantities (e. g. velocity $\vec{v}$ or force $\vec{F}$ from point a to b)

— Now: How do these quantities vary across space? → **field**

— Example: Temperature distribution in a room

- Each point (x, y, z) has an individual temperature $T(x, y, z)$ → **Scalar field**

- Heat flow direction $\vec{\phi}_q(x, y, z)$ at each point → **Vector field**

— Why important for electrical engineering?

- Electric field $\vec{E}(x, y, z)$ → force on charges
- Magnetic field $\vec{B}(x, y, z)$ → force on current

— Note: Because fields are represented by functions, the term 'scalar function' is used as proxy for 'scalar field'; and the term 'vector function' is used for a 'vector field'.

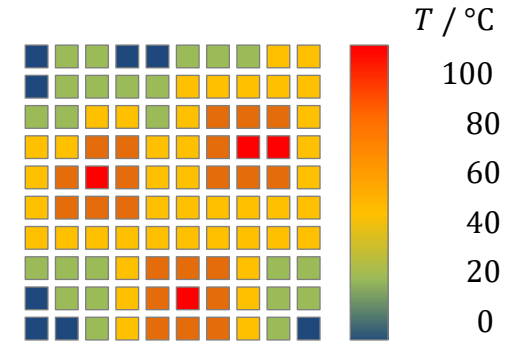


Illustration of a 2D scalar field representing temperature distribution.

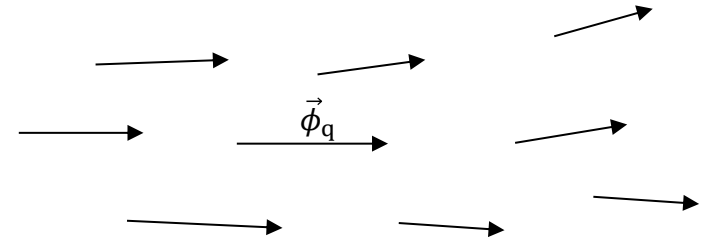


Illustration of a vector field representing heat flow $\vec{\phi}_q$ from left to right.

1.2.1 Scalar and vector fields

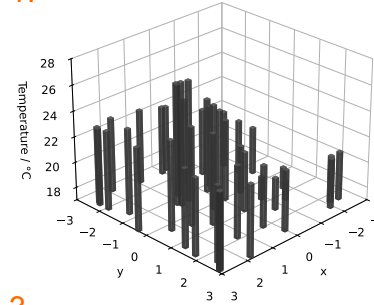
Scalar field

- A scalar field assigns **one scalar value** to each point in space.
- **Mathematical notation:**
 $f: \mathbb{R}^n \rightarrow \mathbb{R}$, where typically $n = 2, 3$
- Engineering Example: **Temperature distribution** $T(x, y)$ on a heat plate (Subfig 1.):
 T is a **scalar quantity** (no direction)
- How scalar fields are **visualized**
 2. **Relief/height surface:** value $\leftrightarrow$ height $z(x, y)$
 3. **Colour map:** value $\leftrightarrow$ colour intensity
 4. **Contour map:** value $\leftrightarrow$ lines of equal value
- From a scalar field $f(x, y, z)$ you can compute the **gradient** $\vec{\nabla} f \rightarrow$ this creates a **vector field**.
- **Example:** $f(x, y, z) = x + 4y + z^2$, find $f(1, 2, 3)$

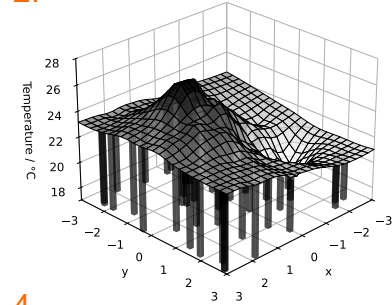
?
Quick check

1. How many dimensions does the field below have?
2. Which visualization do you find most intuitive and why?

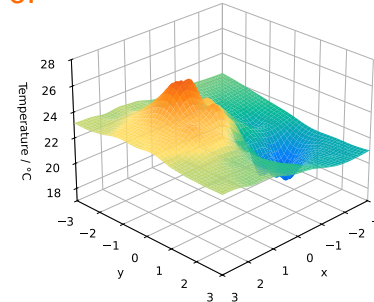
1.



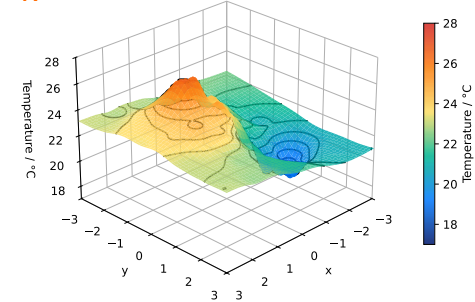
2.



3.



4.

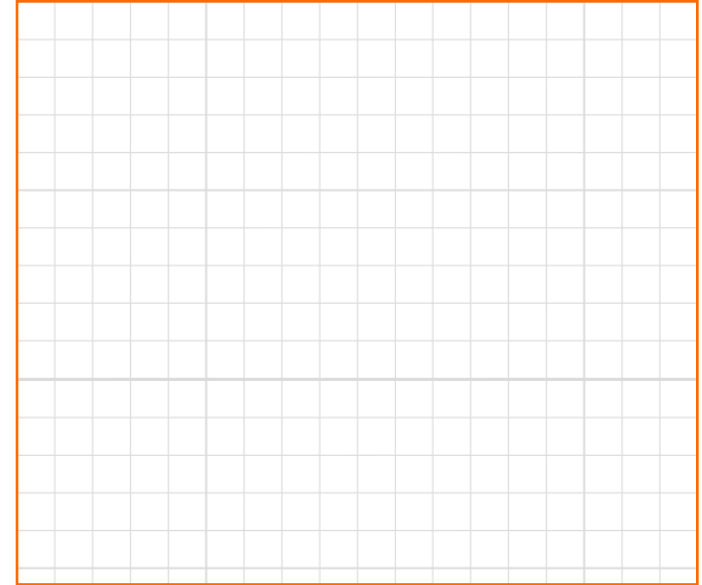
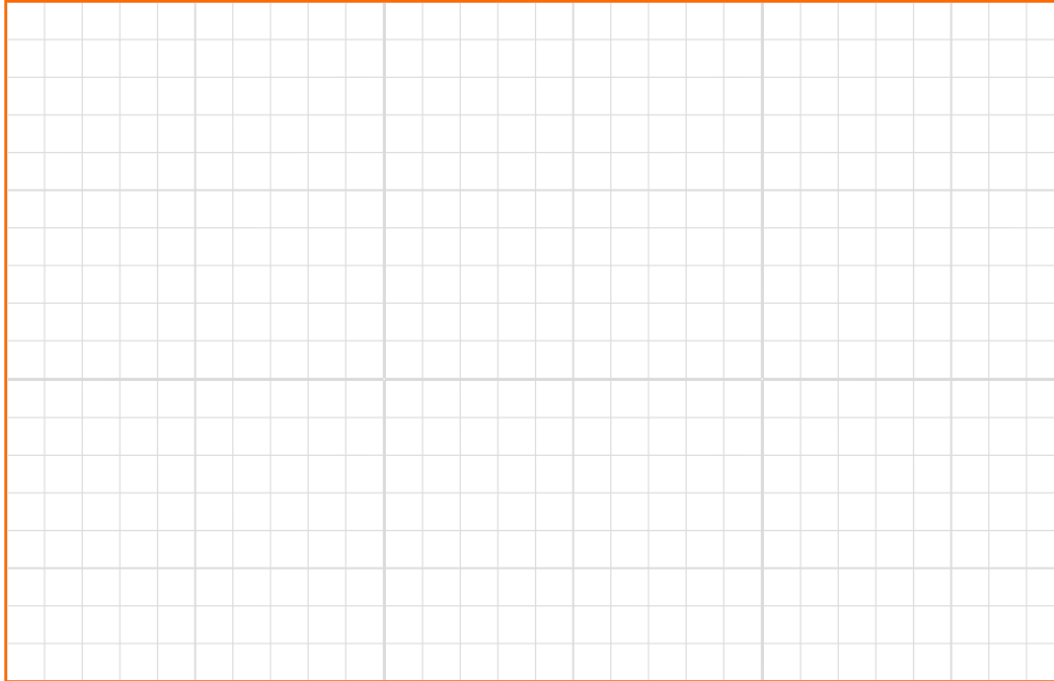


Different ways of visualizing the scalar field $f(x, y) =$
 $22.0 + 4.5 \exp\left(-\frac{(x-0.8)^2 + (y+0.5)^2}{2.0}\right) -$
 $3.5 \exp\left(-\frac{(x+1.2)^2 + (y-0.8)^2}{1.6}\right) + 0.35 x.$

1.2.1 Scalar and vector fields

Exercise: Scalar field

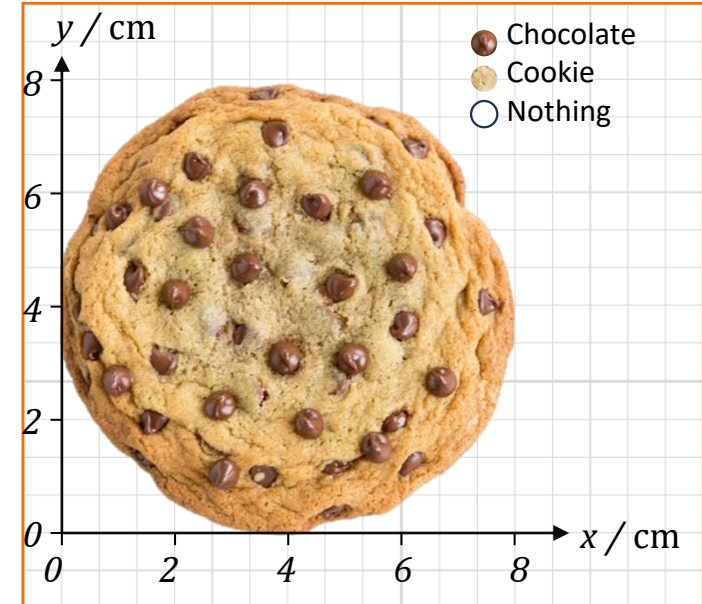
Given is the scalar field $f(x, y) = x^2 + y^2$. Calculate the scalar values for $f(0, 0)$ and $f(3, 3)$. Plot the result. What do you notice?



1.2.1 Scalar and vector fields

Homework: Scalar field

2D Example of a scalar 'cookie field'. Each point is associated to a scalar value (chocolate, cookie, nothing). Evaluate the cookie field at $f(0, 2)$, $f(4, 3)$ and $f(5, 5)$. Identify three arbitrary coordinates for each value.



Mapping of a 2D scalar 'cookie field' to three scalar quantities.

1.2.1 Scalar and vector fields

Vector field

Quick check

1. What is the intuition of the argument in $\vec{v}(x, y, z, t)$?
2. What dependency with respect to the distance r and the polar and azimuthal angle does $\vec{E}(\vec{r})$ have?

- Assigns a **vector** to every point $P \in \mathbb{R}^3$: $\vec{v}: \mathbb{R}^3 \rightarrow \mathbb{R}^3$

Each component is an individual function!

$$\vec{v}(x, y, z) = \begin{bmatrix} v_x(x, y, z) \\ v_y(x, y, z) \\ v_z(x, y, z) \end{bmatrix} = (v_x(x, y, z), v_y(x, y, z), v_z(x, y, z))$$

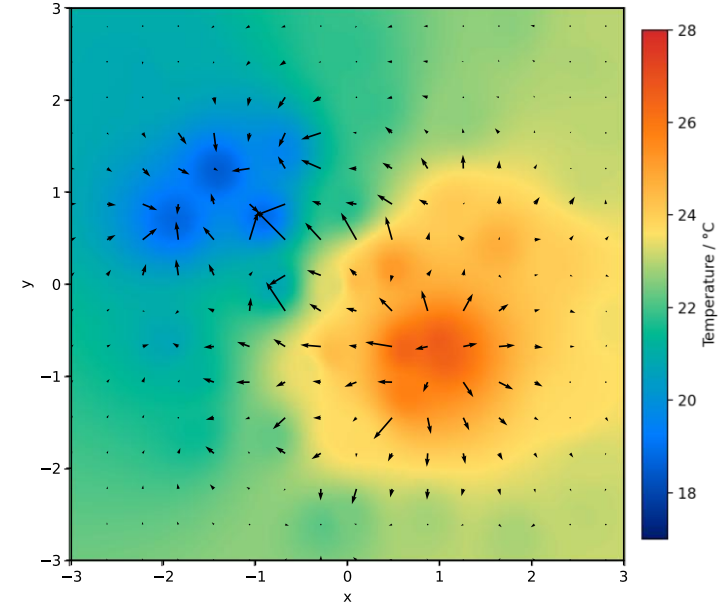
- Illustration as vectors or continuous lines

- Physical examples:

1. **Velocity field:** fluid flow $\vec{v}(x, y, z, t)$
2. **Electric field** of a point charge:

$$\vec{E}(\vec{r}) = \frac{Q}{4 \pi \epsilon_0} \frac{\vec{r}}{\|\vec{r}\|^3}$$

- To evaluate the field at a certain point $P(x, y, z)$, **substitute** x, y, z into each component function v_x, v_y, v_z .
- *Example: Let $\vec{v}(x, y, z) = (4y, 5x)$. Compute $\vec{v}(1, 2, 3)$ and interpret the result.*



A combination of a vector field (arrows) and a scalar field (colour map), visualizing the heat flow from hot to cold areas.

1.2.1 Scalar and vector fields

Exercise: Vector field

Given is the following vector field $\vec{v}(x, y) = \begin{bmatrix} y^2 \\ 0.5x + y + 1 \end{bmatrix}$. Calculate the displacement vectors $\vec{v}_1(0, 0)$, $\vec{v}_2(1, 1)$, $\vec{v}_3(2, 2)$. Calculate the end point of these vectors and plot the result.



1.2.1 Scalar and vector fields

Check: Vector vs. scalar vs. vector field vs. scalar field

Categorize the following as **scalar**, **scalar field**, **vector** or **vector field**: **1.** Electric field; **2.** $(3,4,6)$; **3.** Altitude of a mountain top; **4.** $(3,4, x, 6)$; **5.** Electric potential; **6.** $2x + 3y^2$; **7.** Map of the alps; **8.** Steepest path uphill **9.** battery state of charge. Suggest other examples describing one of the above.



Summary

- Scalar field: $f: \mathbb{R}^n \rightarrow \mathbb{R}$.
- Vector field: $\vec{F}: \mathbb{R}^n \rightarrow \mathbb{R}^n$.
- *Example*

$$\begin{aligned} f(x, y, z) &= 2x + 3xy + z^2 && \rightarrow f(1, 2, 3) = 2 \cdot 1 + 3 \cdot 1 \cdot 2 + 3^2 = 17 \\ \vec{F}(x, y, z) &= \begin{bmatrix} 2x + 3xy + z^2 \\ x^2 + 3z \\ x^2 + y^2 \end{bmatrix} && \rightarrow \vec{F}(1, 2, 3) = \begin{bmatrix} 2 \cdot 1 + 3 \cdot 1 \cdot 2 + 3^2 \\ 1^2 + 3 \cdot 3 \\ 1^2 + 2^2 \end{bmatrix} = \begin{bmatrix} 17 \\ 10 \\ 5 \end{bmatrix} \end{aligned}$$

1.2.1 Scalar and vector fields

1.2.2 The del operator

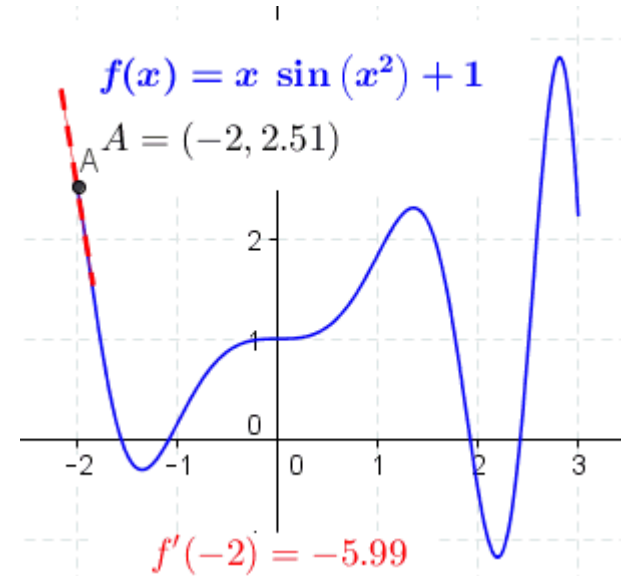
1.2.2 The del operator

Partial derivative



If you can differentiate single-variable functions, you can compute partial derivatives too.

- For a function of **one variable** $f(x)$, the **derivative** $\frac{df}{dx}$ gives the **slope** (rate of change w. r. t. x).
- For a function of **two variables** $f(x, y)$, you must specify **which variable changes, and which is constant**.
 - ┐ Vary x , hold y constant: Partial derivative w. r. t. x
 - ┐ Vary y , hold x constant: Partial derivative w. r. t. y
- **Partial-derivative symbol:** ∂ ('curly d', read 'partial')
- Example: Let $f(x, y) = (x^2 + 2y)$, then
$$\frac{\partial}{\partial x} f(x, y) = 2x$$
$$\frac{\partial}{\partial y} f(x, y) = 2$$
- **Note:** There is also the **total derivative**, used when variables are not independent (e. g., $y = y(x)$); it gives the rate of change of $f(x, y(x))$ with respect to x (chain rule).

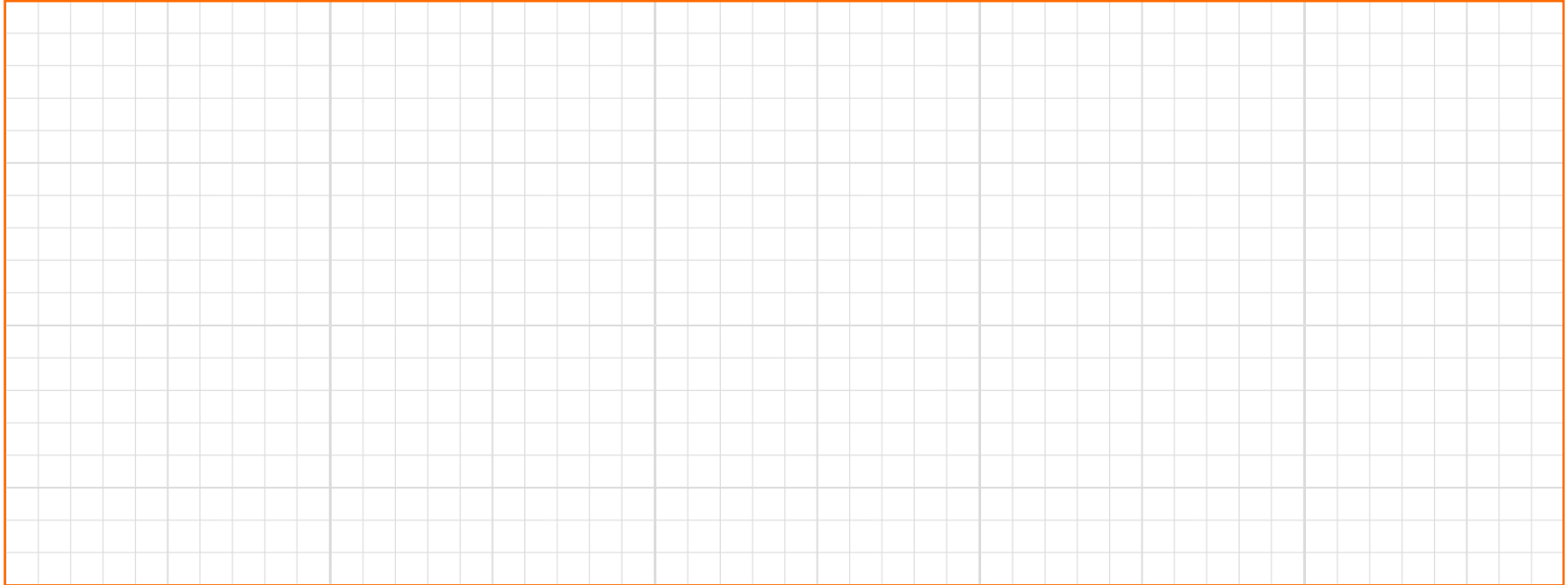


The derivative can be considered the slope of the tangent line of a function.

1.2.2 The del operator

Example: Partial derivative

Let $f(x, y, z) = (3x + y^{-2} + 2x^2z^4 + y^3e^{2z})$. Calculate $\frac{\partial}{\partial x} f(x, y, z)$.



1.2.2 The del operator

Homework: Partial derivative

Let $f(x, y, z) = (3x + y^{-2} + 2x^2z^4 + y^3e^{2z})$. Calculate $\frac{\partial}{\partial y}f(x, y, z)$ and $\frac{\partial}{\partial z}f(x, y, z)$



1.2.2 The del operator

Del or nabla operator

- The del operator $\vec{\nabla}$ is a **differential operator**
- In Cartesian coordinates, its components are partial derivatives.

$$\vec{\nabla} = \vec{e}_x \frac{\partial}{\partial x} + \vec{e}_y \frac{\partial}{\partial y} + \vec{e}_z \frac{\partial}{\partial z}$$

- Del **bundles several key differential operations** in vector calculus:

- 7 **Gradient**: direction of steepest change:

$$\text{grad } f = \vec{\nabla} f$$

- 7 **Divergence**: sources and sinks

$$\text{div } \vec{F} = \vec{\nabla} \cdot \vec{F}$$

- 7 **Rotation** / curl : rotation and circulation

$$\text{curl } \vec{F} = \text{rot } \vec{F} = \vec{\nabla} \times \vec{F}$$

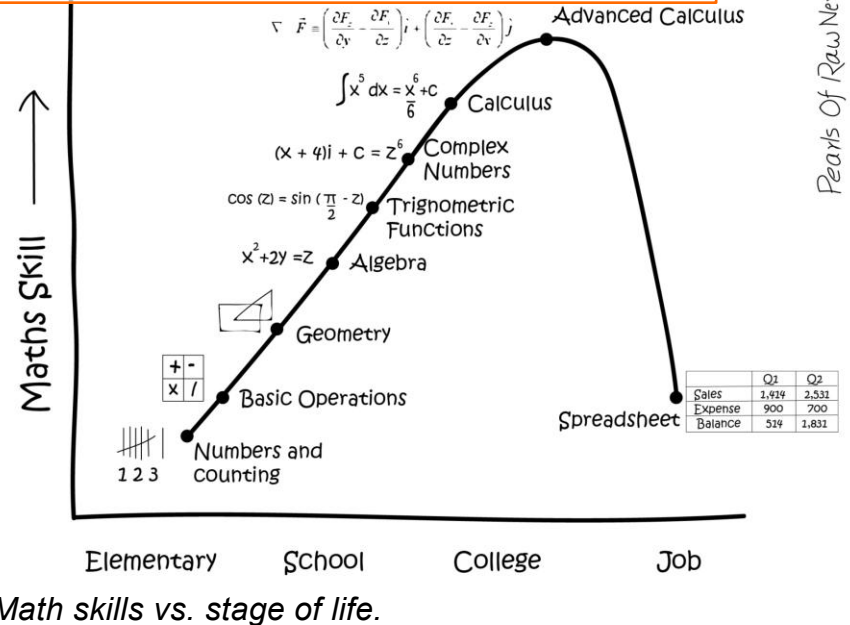
Common in US

Common in EU

? Can you identify the operations (formal products) corresponding to $\vec{\nabla} f$ (grad),
Quick check $\vec{\nabla} \cdot \vec{F}$ (div) and $\vec{\nabla} \times \vec{F}$ (rot)?



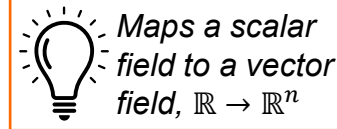
Del, also known as 'nabla' and written as $\vec{\nabla}$ (arrow for better intuition as formal vector), is the 'Swiss Army knife' of field analysis. Learn its behaviour and calculation rules thoroughly!



Pearls Of Raw Newdism

1.2.2 The del operator

Gradient



- The gradient **points in the direction of steepest increase** of a scalar field.
- $\text{grad } f$ is a **vector field of first derivatives** of a **scalar field** f

$$\text{grad } f(x_1, \dots, x_n) = \vec{\nabla} f(x_1, \dots, x_n) = \left(\frac{\partial f}{\partial x_1}, \dots, \frac{\partial f}{\partial x_n} \right)$$

- In Cartesian coordinates:

vector operator $\vec{\nabla} f(x, y, z) = \left(\frac{\partial f}{\partial x}, \frac{\partial f}{\partial y}, \frac{\partial f}{\partial z} \right) = \frac{\partial f}{\partial x} \vec{e}_x + \frac{\partial f}{\partial y} \vec{e}_y + \frac{\partial f}{\partial z} \vec{e}_z$

scalar field vector field

- Example: Calculate the gradient of Filbert Street, SF.

$$= \frac{\Delta z}{\Delta x} x + \frac{\Delta z}{\Delta y} y = \frac{3}{10} x + \frac{0}{\Delta y} \cdot y = 0.3 x$$



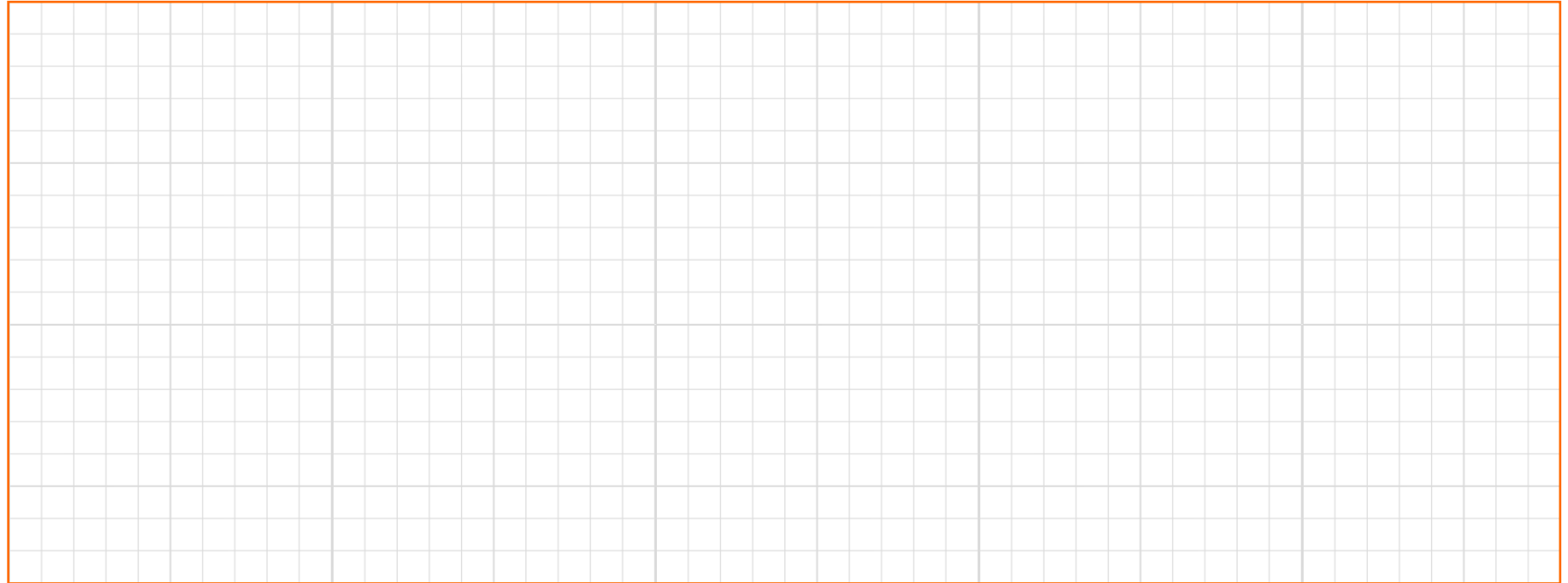
Filbert Street, San Francisco ($m = 31.5 \%$).

1.2.2 The del operator

Example: Gradient in cartesian coordinates

EXAM	Typical
• —	Exam
• —	Question
• —	

Given is the function $f(x, y, z) = 3x + x^2 y z + y z^3$. Please calculate the gradient and the slope of the function at the points $P_1(0, 0, 0)$ and $P_2(1, 5, 10)$.



1.2.2 The del operator

Gradient in other coordinate systems

- Gradient in **spherical coordinate** system

$$\vec{\nabla}f(r, \theta, \varphi) = \left(\frac{\partial f}{\partial r}, \frac{1}{r} \frac{\partial f}{\partial \theta}, \frac{1}{r \sin \theta} \frac{\partial f}{\partial \varphi} \right) \\ = \frac{\partial f}{\partial r} \vec{e}_r + \frac{1}{r} \frac{\partial f}{\partial \theta} \vec{e}_\theta + \frac{1}{r \sin \theta} \frac{\partial f}{\partial \varphi} \vec{e}_\varphi$$

⌊ With the **radial distance** r , the **polar angle** θ and the **azimuthal angle** φ .

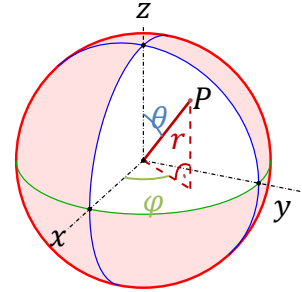
⌊ Typical use case: electrical fields of **point charges**.

- Gradient in **cylindrical coordinate** system

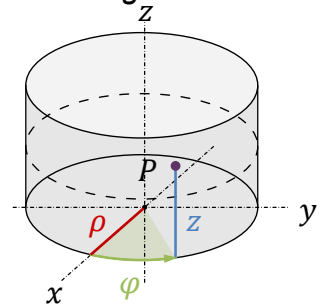
$$\vec{\nabla}f(\rho, \varphi, z) = \left(\frac{\partial f}{\partial \rho}, \frac{1}{\rho} \frac{\partial f}{\partial \varphi}, \frac{\partial f}{\partial z} \right) = \frac{\partial f}{\partial \rho} \vec{e}_\rho + \frac{1}{\rho} \frac{\partial f}{\partial \varphi} \vec{e}_\varphi + \frac{\partial f}{\partial z} \vec{e}_z$$

⌊ With the **radial distance** ρ , the **height** z and the **angular coordinate** φ .

⌊ Typical use case: electrical fields around line charges, i. e. current carrying **wires**.



Spherical coordinates (r, θ, φ) as commonly used around point charges.



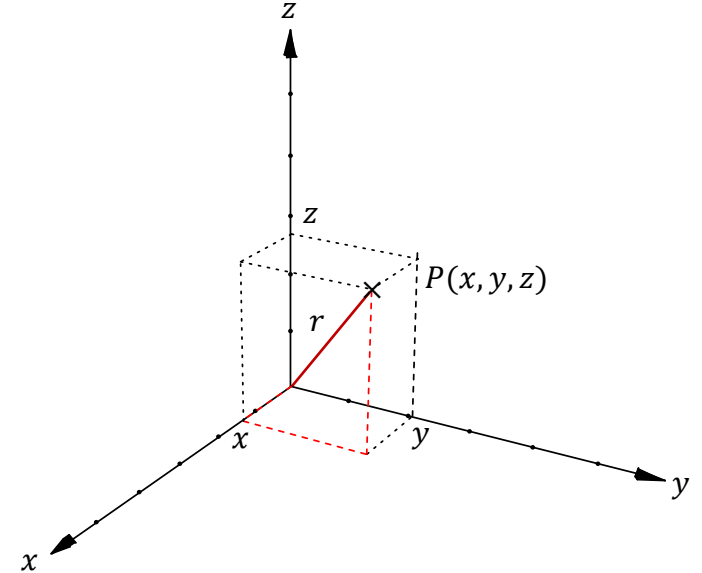
Cylindrical coordinates (ρ, φ, z) as commonly used around wires.

1.2.2 The del operator

Example: Electric field of a point charge using cartesian coordinates

EXAM	Typical
• —	Exam
• —	Question
• —	
• —	

Calculate the electrical field $\vec{E} = -\vec{\nabla} \phi$ of a point charge Q at a point $P(x, y, z)$ at the distance r using cartesian coordinates. The potential of a point charge is given by $\phi(r) = \frac{Q}{4\pi\epsilon_0} \frac{1}{r}$.



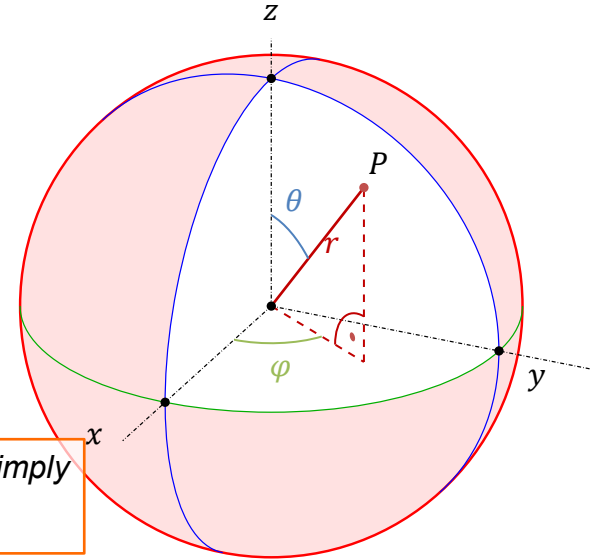
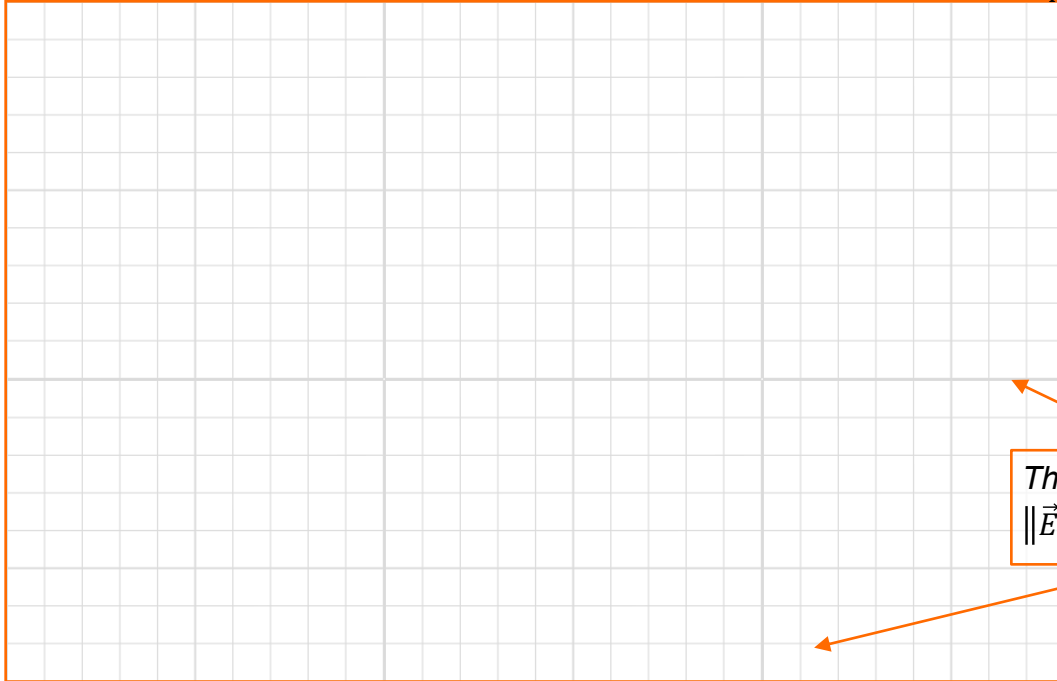
The point P at a distance $r = \sqrt{x^2 + y^2 + z^2}$ from the origin.

1.2.2 The del operator

Example: Electric field of a point charge using polar coordinates

EXAM	Typical
•	Exam
•	Question
•	
•	

Calculate the electrical field $\vec{E} = -\vec{\nabla} \phi$ of a point charge Q at a point $P(x, y, z)$ at the distance r using spherical coordinates. The potential of a point charge is given by $\phi(r) = \frac{Q}{4 \pi \epsilon_0 r}$.



The norm is simply
$$\|\vec{E}\| = \frac{Q}{4 \pi \epsilon_0 r^2}$$

Spherical coordinates (r, θ, φ) as commonly used around point charges.

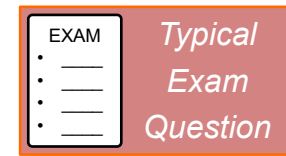
1.2.2 The del operator

Directional derivative

- Slope from a point (x, y, z) in an arbitrary direction $\vec{v}$.
- Calculated from the **scalar product** of the normalized **direction vector** $\frac{\vec{v}}{\|\vec{v}\|}$ by the **gradient** of the function at (x, y, z) :

$$\frac{\partial f(x, y, z)}{\partial \vec{v}} = \frac{\vec{v}}{\|\vec{v}\|} \cdot \vec{\nabla} f(x, y, z)$$

- *Example: Calculate the directional derivative when walking up Filbert street diagonally from $(0,0,0)$ to $(10,5,3)$.*



*Filbert Street, San Francisco ($m = 31.5\%$).
The vector points diagonally from $(0,0,0)$ to $(10,5,3)$.*

1.2.2 The del operator

Important concept: Circulation

- **Circulation** Γ measures the net “along-the-path” component of a vector field $\vec{F}$ around a closed loop.

- For a closed, oriented curve $\partial\Sigma$ in a vector field $\vec{F}$

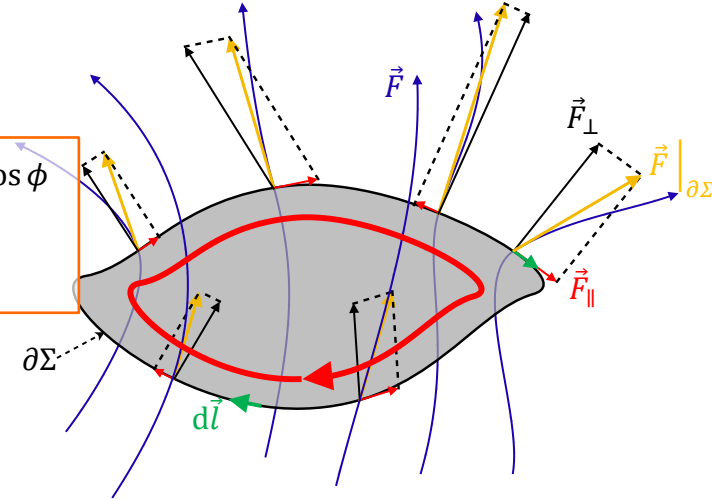
$$\Gamma = \oint_{\partial\Sigma} \vec{F} \cdot d\vec{l}$$

$$\begin{aligned}\vec{F} \cdot d\vec{l} &= \|\vec{F}\| \|d\vec{l}\| \cos \phi \\ \vec{F}_{\perp} \cdot d\vec{l} &= 0 \\ \vec{F}_{\parallel} \cdot d\vec{l} &= \|\vec{F}_{\parallel}\| \|d\vec{l}\|\end{aligned}$$

- ⌈ $d\vec{l}$: infinitesimal tangent vector along $\partial\Sigma$
- ⌈ $\vec{F} \cdot d\vec{l}$: projects $\vec{F}$ onto the tangent direction
- ⌈ The integral adds these contributions around the loop
- **Physical Intuition**
 - ⌈ $\Gamma > 0$: On average, the tangential component points along the direction of traversal (“tailwind”).
 - ⌈ $\Gamma < 0$: On average, the tangential component opposes the direction of traversal (“headwind”).
 - ⌈ $\Gamma = 0$: No net circulation.
- **Orientation Matters:** Reversing the direction of traversal flips the sign: $\Gamma \rightarrow -\Gamma$.

?
Quick
check

1. Name a real-world situation with **nonzero circulation** $\Gamma \neq 0$. What would make $|\Gamma|$ larger/smaller?
2. Name a real-world situation with $\Gamma = 0$ for a closed loop. Explain the cancellation in one sentence.
3. What is the key field-property difference between 1. and 2.?



Circulation around a closed curve $\partial\Sigma$.

 In chapter 3 we'll learn more about $\oint \vec{E} \cdot d\vec{l}$ (induction) and $\oint \vec{H} \cdot d\vec{l}$ (source of magnetic fields)

1.2.2 The del operator

Curl or rotation



2D: Maps a vector field to a scalar, $\mathbb{R}^2 \rightarrow \mathbb{R}$

3D: Maps a vector field to a vector field, $\mathbb{R}^3 \rightarrow \mathbb{R}^3$

More dimensions: **standard vector calculus fails**

- **Curl** measures the **infinitesimal circulation** of a vector field $\vec{F}$, i. e. the **circulation per unit area around a point**

- Mathematical definition

$$\text{rot } \vec{F} = \vec{\nabla} \times \vec{F}$$

Different ways of representing $\text{rot } \vec{F}$ in Cartesian coordinates.

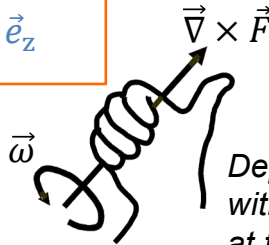
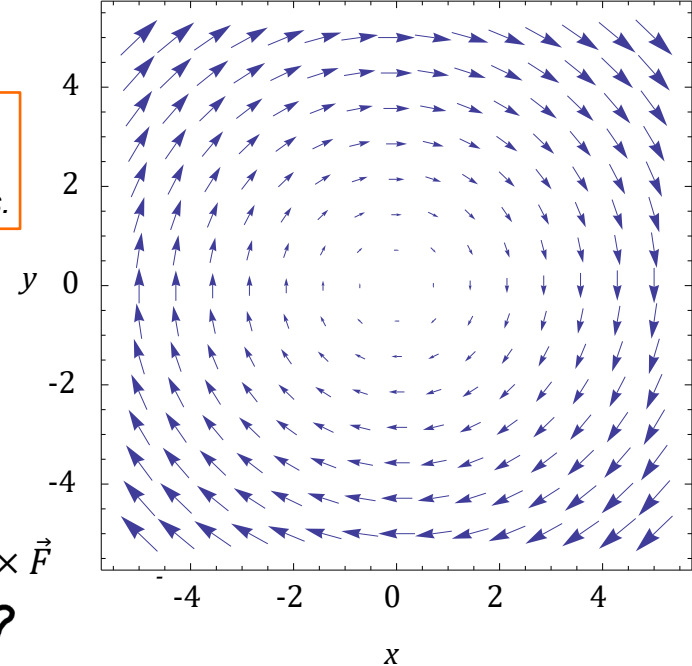
- In Cartesian coordinates:

$$\begin{aligned} \vec{\nabla} \times \vec{F} &= \begin{bmatrix} \frac{\partial}{\partial x} \\ \frac{\partial}{\partial y} \\ \frac{\partial}{\partial z} \end{bmatrix} \times \begin{bmatrix} F_x \\ F_y \\ F_z \end{bmatrix} = \det \begin{bmatrix} \vec{e}_x & \vec{e}_y & \vec{e}_z \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ F_x & F_y & F_z \end{bmatrix} = \begin{bmatrix} \vec{e}_x & \vec{e}_y & \vec{e}_z \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ F_x & F_y & F_z \end{bmatrix} \\ &= \left(\frac{\partial F_z}{\partial y} - \frac{\partial F_y}{\partial z} \right) \vec{e}_x + \left(\frac{\partial F_x}{\partial z} - \frac{\partial F_z}{\partial x} \right) \vec{e}_y + \left(\frac{\partial F_y}{\partial x} - \frac{\partial F_x}{\partial y} \right) \vec{e}_z \end{aligned}$$

- Physical examples

- ⌈ $\text{rot } \vec{F} \neq 0$: rotation: magnetic field around wire
- ⌈ $\text{rot } \vec{F} = 0$: field is 'irrotational': electrostatic field

- **Note: Right-fist rule to find the direction of rotation.**



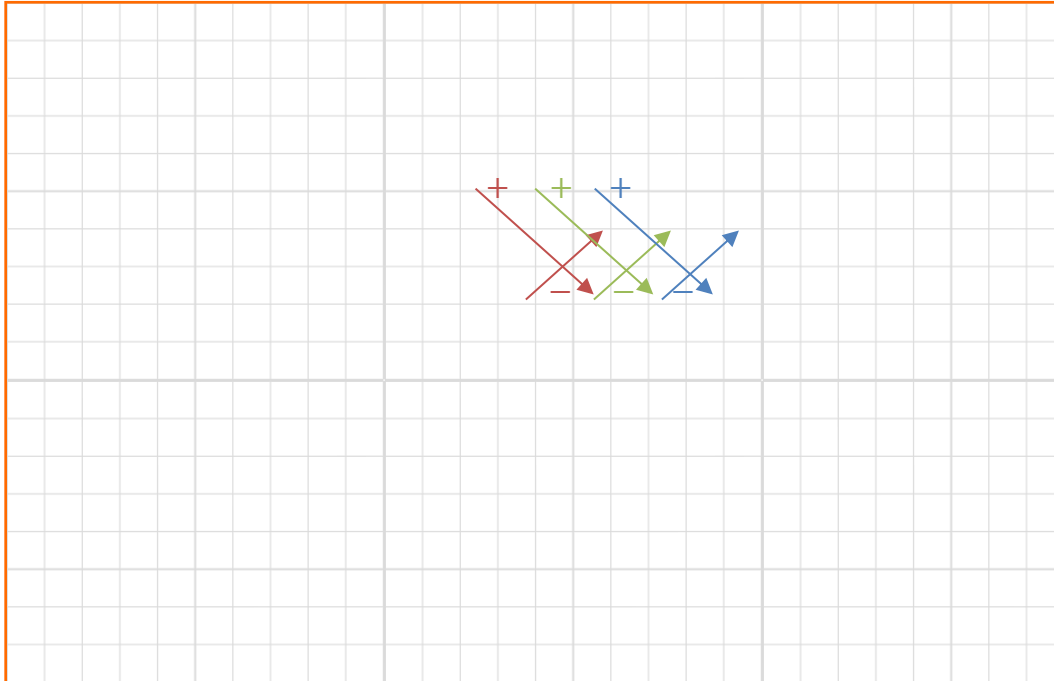
Depiction of a two-dimensional vector field with a uniform curl. Each arrow shows $\vec{v}(x, y)$ at that point.

1.2.2 The del operator

Example: Curl 1/2

EXAM	Typical
•	Exam
•	Question
•	
•	

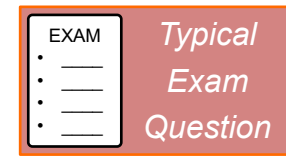
Calculate the curl of $\vec{F}(x, y, z) = (y, -x, 0)$. In which direction does the curl rotate? Does this field also diverge?



Vector field $\vec{F}(x, y) = (y, -x, 0)$ (**left**) and its curl (**right**).

1.2.2 The del operator

Example: Curl 2/2



Calculate the curl of $\vec{F}(x, y, z) = (0, x^2, 0)$. In which direction does the field rotate? Does this field also diverge?



Place a virtual stick rotating around its center to test if the field is rotating at this point.

Vector field $\vec{F}(x, y) = (0, x^2, 0)$ (**left**) and its curl (**right**).

1.2.2 The del operator

Important concept: Flux

?
Quick
check

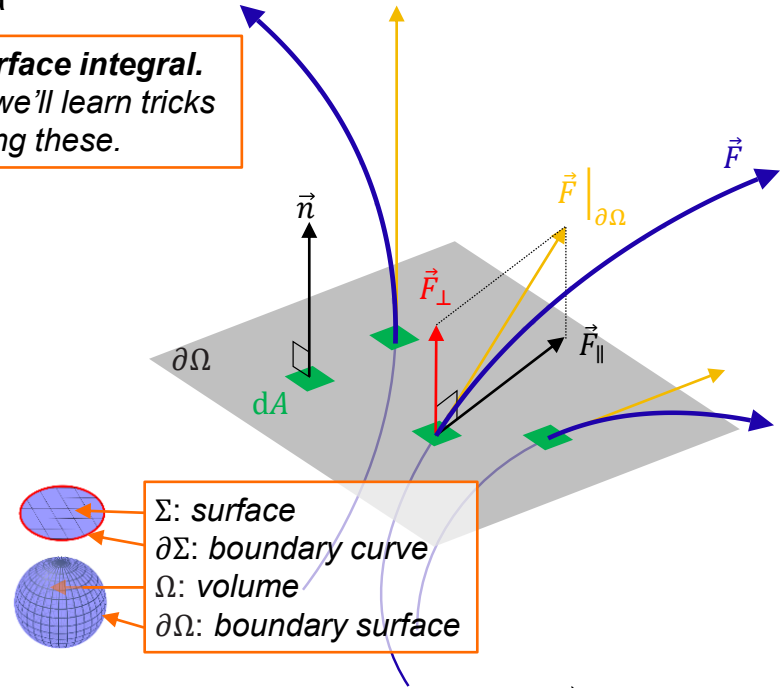
1. Name a real-world situation where mass flows through a surface. What surface? What direction?
2. Pick a different flux (heat, electric, or magnetic). In one sentence: What flows through what surface? Share with a peer.

- Flux Φ describes the net flow of a **vector field** $\vec{F}$ through a **closed surface** $\partial\Omega$:

$$\Phi = \oiint_{\partial\Omega} \vec{F} \cdot \vec{n} \, dA = \oiint_{\partial\Omega} \vec{F} \cdot d\vec{A}$$

This is a **surface integral**.
No worries we'll learn tricks circumventing these.

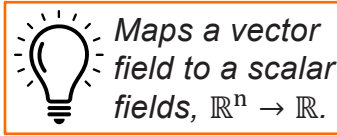
- $\vec{n} \, dA$: infinitesimal oriented area element on surface $\partial\Omega$
- $\vec{F} \cdot \vec{n}$: projection of $\vec{F}$ onto the local outward normal $\vec{n}$
- The integral adds these contributions over the surface
- Examples
 - Velocity fields: **mass flux** $\rho \oiint \vec{v} \cdot \vec{n} \, dA$
 - Electric fields: **electric flux** (Gauss' law) $\oiint \vec{E} \cdot \vec{n} \, dA$
 - Magnetic fields: **magnetic flux** $\oiint \vec{B} \cdot \vec{n} \, dA$
 - 'Flux of cars' through a tunnel on single lane side street vs. six lane Autobahn.



The field lines of a vector field $\vec{F}$ through a surface Σ with unit normal $\vec{n}$.

1.2.2 The del operator

Divergence



- **Divergence** measures the net outward **flux density** at a point:
 - 7 $\text{div } \vec{F} > 0$: **source** (field lines emerge) → e. g., positive charge
 - 7 $\text{div } \vec{F} < 0$: **sink** (field lines converge) → e. g., negative charge
 - 7 $\text{div } \vec{F} = 0$: **no net flow** in/out at this point (no net field or field lines in closed loops)
- A field with **divergence zero at all points** is called **divergence-free vector field**
- Mathematically a **dot product of $\vec{\nabla}$ with vector field $\vec{F}$** :

$$\text{div } \vec{F} = \vec{\nabla} \cdot \vec{F} = \left(\frac{\partial}{\partial x_1}, \dots, \frac{\partial}{\partial x_n} \right) \cdot (F_1, \dots, F_n)$$

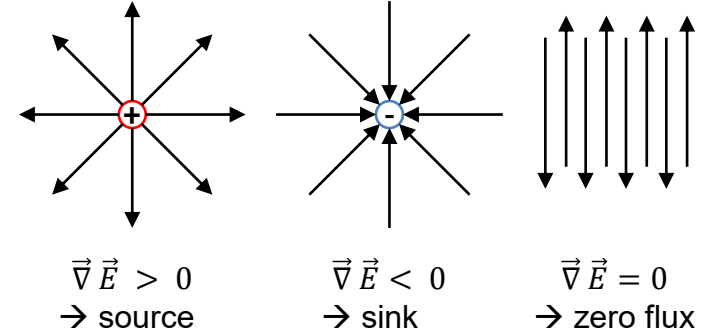
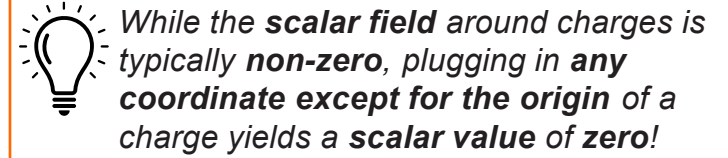
- In cartesian coordinates:

vector operator

$\vec{\nabla} \cdot \vec{F} = \frac{\partial F_x}{\partial x} + \frac{\partial F_y}{\partial y} + \frac{\partial F_z}{\partial z}$

scalar field

vector field



Application of the div operator for the electric field $\vec{E}$.

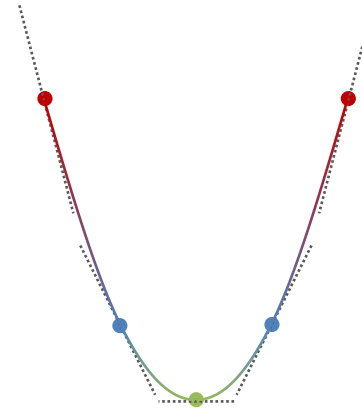
1.2.2 The del operator

Example: Divergence of a velocity field

EXAM	Typical
•	Exam
•	Question
•	
•	

You are given the vector field $\vec{v} = (x^2, 0)$. Please calculate $\text{div}(\vec{v})$ and draw both the vector field and the scalar field at $P_1(-2,0)$, $P_2(-1,1)$, $P_3(0,1)$, $P_4(1,1)$, $P_5(1,2)$, $P_6(2,0)$. How can you describe the field and the div?

- Speed/velocity in positive x direction.
- Velocity field illustrated by vectors
- Vector norm illustrated by colour map



- $\text{div}(\vec{v}) = \vec{\nabla} \cdot (x^2, 0) = 2x$
- Plug in at given points
- $\vec{\nabla} \cdot \vec{v}_1(-2,0) = -4$
- $\vec{\nabla} \cdot \vec{v}_2(-1,1) = -2$
- $\vec{\nabla} \cdot \vec{v}_3(0,1) = 0$
- $\vec{\nabla} \cdot \vec{v}_4(1,1) = 2$
- $\vec{\nabla} \cdot \vec{v}_5(1,2) = 2$
- $\vec{\nabla} \cdot \vec{v}_6(2,1) = 4$
- Tangent to each point

- Regions acting as source, sink or without flow
- Physical interpretation
 - $\text{div}(\vec{v}) < 0 \triangleq \text{Break}$
 - $\text{div}(\vec{v}) = 0 \triangleq \text{Stopp}$
 - $\text{div}(\vec{v}) > 0 \triangleq \text{Accelerate}$

1.2.2 The del operator

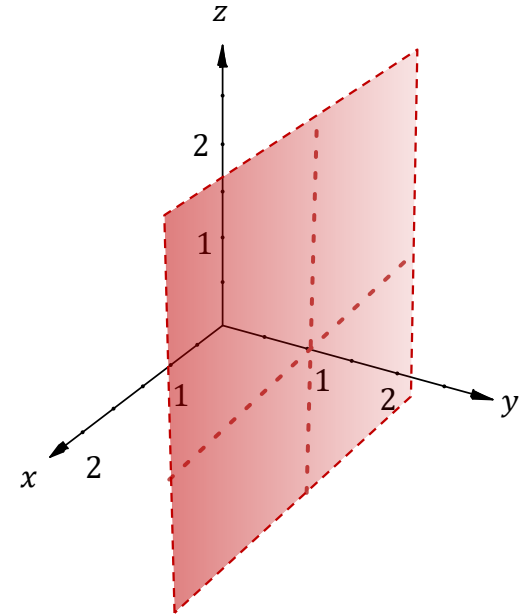
Example: Visualize divergence

Calculate the divergence of $\vec{F}(x, y, z) = (-xy, -x, z)$

Where do you find zero flux? Describe the object.



EXAM	Typical
•	Exam
•	Question
•	
•	



Plane of zero flux for the vector field $\vec{F}(x, y, z)$.

1.2.2 The del operator

The fundamental calculus theorems of electromagnetism

- **Stokes Theorem** (line integral $\leftrightarrow$ surface integral)

$$\oint_{\partial\Sigma} \vec{F} \cdot d\vec{l} = \iint_{\Sigma} (\vec{\nabla} \times \vec{F}) \cdot \vec{n} dA$$

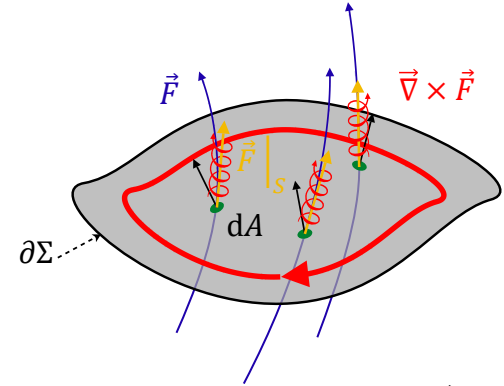
Circulation around a boundary curve = flux of curl through the enclosed surface

- **Divergence (Gauss) Theorem** (surface integral $\leftrightarrow$ volume integral)

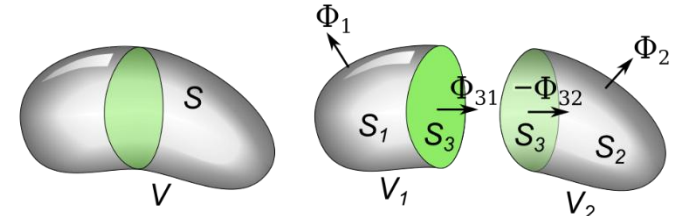
$$\oiint_{\partial\Omega} \vec{F} \cdot \vec{n} dA = \iiint_{\Omega} (\vec{\nabla} \cdot \vec{F}) dV$$

Net flux through a closed surface = sum of divergence (sources) inside the volume

- **Key Idea:** Convert difficult *boundary integrals* into simpler *interior integrals* using curl/div operators.
- **Why this matters:** You'll use these to move between integral and differential forms of **Maxwell's equations** (Faraday's law, Ampère's law, Gauss's law).



*Circulation around a closed curve $\vec{C}$ is equivalent to the **curl of the field** integrated over the surface S spanned by curve $\vec{C}$*



A volume divided into two sub-volumes. At right, the two sub-volumes are separated to show the flux out of the different surfaces.

1.2.2 The del operator

Looking forward: Maxwell's equations in equilibrium

Integral form	Differential form	Name and physical meaning
I. $\oiint_{\partial\Omega} \vec{D} \cdot \vec{n} \, dA = Q_{\text{encl}}$	$\vec{\nabla} \cdot \vec{D} = \rho$	Gauss' law: Net electric flux through a closed surface equals the enclosed charge.
II. $\oint_{\partial\Sigma} \vec{E} \cdot d\vec{l} = 0$	$\vec{\nabla} \times \vec{E} = 0$	Conservative electric field: Electrostatic fields have zero curl; they derive from a scalar potential $\vec{E} = -\vec{\nabla} \varphi$.
III. $\oiint_{\partial\Omega} \vec{B} \cdot \vec{n} \, dA = 0$	$\vec{\nabla} \cdot \vec{B} = 0$	Gauss's law (magnetic): magnetic field is solenoidal (divergence-free); field lines form closed loops.
IV. $\oint_{\partial\Sigma} \vec{H} \cdot d\vec{l} = I$	$\vec{\nabla} \times \vec{H} = \vec{j}$	Ampère's law (static): Electric current produces a magnetic field (right-hand rule: thumb = current, fingers = $\vec{H}$).

Summary

- Del $\vec{\nabla}$ is a universal operator.
- When performing calculations you can handle $\vec{\nabla}$ just as a vector.
- Three vector operations (scalar multiplication, dot product and cross product) yield three different outcomes:
 - ▮ $\text{grad } f = \vec{\nabla} f$ is a vector field.
 - ▮ $\text{div } \vec{F} = \vec{\nabla} \cdot \vec{F}$ is a scalar field (net outflow density).
 - ▮ $\text{curl } \vec{F} = \vec{\nabla} \times \vec{F}$ is a vector field (local rotation/circulation density).

1.2.2 The del operator

Calculation rules

More exercises

Appendix to 1.2.2: The del operator

Appendix to 1.2.2: The del operator

Calculation rules

f and g are scalars.
 $\vec{u}$ and $\vec{v}$ are vectors.

Operators are linear

- $\text{grad}(f + g) = \text{grad } f + \text{grad } g$ and $\text{grad}(\lambda f) = \lambda \text{grad } f$ for $\lambda \in \mathbb{R}$
- $\text{div}(\vec{u} + \vec{v}) = \text{div } \vec{u} + \text{div } \vec{v}$ and $\text{div}(\lambda \vec{v}) = \lambda \text{div } \vec{v}$ for $\lambda \in \mathbb{R}$
- $\text{rot}(\vec{u} + \vec{v}) = \text{rot } \vec{u} + \text{rot } \vec{v}$ and $\text{rot}(\lambda \vec{v}) = \lambda \text{rot } \vec{v}$ for $\lambda \in \mathbb{R}$

Product rule

- $\text{grad}(f \cdot g) = f \text{grad } g + g \text{grad } f$
- $\text{grad}(\vec{u} \cdot \vec{v}) = (\text{div } \vec{v}) \vec{u} + (\text{div } \vec{u}) \vec{v}$
- $\text{div}(f \vec{v}) = f \text{div } \vec{v} + (\text{grad } f) \cdot \vec{v}$
- $\text{rot}(f \vec{v}) = f \text{rot } \vec{v} + (\text{grad } f) \times \vec{v}$
- $\text{div}(\vec{u} \times \vec{v}) = -\vec{u} \cdot \text{rot } \vec{v} + \vec{v} \cdot \text{rot } \vec{u}$

Repeated application

- $\text{div}(\text{grad } f) = \Delta f$ (Laplace-Operator)
- $\text{rot}(\text{grad } f) = \vec{0}$ (Scalar potentials are conservative and do not rotate)
- $\text{div}(\text{rot } \vec{v}) = 0$ (Eddy fields have zero flux)

Appendix to 1.2.2: The del operator

Calculation rules in the del operator notation

Operators are **linear**

- $\vec{\nabla}(f + g) = \vec{\nabla}f + \vec{\nabla}g$ and $\vec{\nabla}(\lambda f) = \lambda \vec{\nabla}f$ for $\lambda \in \mathbb{R}$
- $\vec{\nabla}(\vec{u} + \vec{v}) = \vec{\nabla}\vec{u} + \vec{\nabla}\vec{v}$ and $\vec{\nabla}(\lambda \vec{v}) = \lambda \vec{\nabla} \cdot \vec{v}$ for $\lambda \in \mathbb{R}$
- $\vec{\nabla} \times (\vec{u} + \vec{v}) = \vec{\nabla} \times \vec{u} + \vec{\nabla} \times \vec{v}$ and $\vec{\nabla} \times (\lambda \vec{v}) = \lambda \vec{\nabla} \times \vec{v}$ for $\lambda \in \mathbb{R}$

Product rule

- $\vec{\nabla}(f \cdot g) = f \cdot \vec{\nabla}g + g \vec{\nabla}f$
- $\vec{\nabla}(\vec{u} \cdot \vec{v}) = (\vec{\nabla}\vec{v})\vec{u} + (\vec{\nabla}\vec{u})\vec{v}$
- $\vec{\nabla}(f\vec{v}) = f\vec{\nabla}\vec{v} + (\vec{\nabla}f)\vec{v}$
- $\vec{\nabla} \times (f\vec{v}) = f\vec{\nabla} \times \vec{v} + (\vec{\nabla}f) \times \vec{v}$
- $\vec{\nabla}(\vec{u} \times \vec{v}) = -\vec{u} \cdot \vec{\nabla} \times \vec{v} + \vec{v} \cdot \vec{\nabla} \times \vec{u}$

Repeated application

- $\vec{\nabla}(\vec{\nabla}f) = \Delta f$ (Laplace-Operator)
- $\vec{\nabla} \times (\vec{\nabla}f) = \vec{0}$ (Scalar potentials are conservative and do not rotate)
- $\vec{\nabla}(\vec{\nabla} \times \vec{v}) = \vec{0}$ (Eddy fields have zero flux)

Appendix to 1.2.2: The del operator

Overview

Operation	Notation	Description
Del, or Nabla operator	$\vec{\nabla} = \left(\frac{\partial}{\partial x_1}, \dots, \frac{\partial}{\partial x_n} \right)$	Partial derivative of the component x_i
Laplace operator	$\Delta = \vec{\nabla}^2 = \left(\frac{\partial^2}{\partial x_1^2}, \dots, \frac{\partial^2}{\partial x_n^2} \right)$	Second partial derivative of x_i , maps between scalar fields.
Gradient	$\text{grad}(f) = \vec{\nabla} f$	Maps a scalar field to a vector field, $\mathbb{R} \rightarrow \mathbb{R}^n$
Divergence	$\text{div}(\vec{v}) = \vec{\nabla} \cdot \vec{v}$	Maps a vector field to a scalar field, $\mathbb{R}^n \rightarrow \mathbb{R}$
Rotor (also: curl)	$\text{rot}(\vec{v}) = \vec{\nabla} \times \vec{v}$	Maps a vector field to a vector field, $\mathbb{R}^3 \rightarrow \mathbb{R}^3$

Appendix to 1.2.2: The del operator

Homework : Curl of gradient is zero

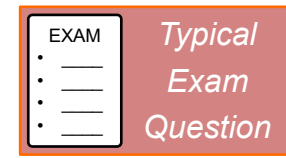
EXAM	Typical Exam Question
• —	
• —	
• —	

Demonstrate that $\vec{\nabla} \times (\vec{\nabla} f) = \vec{0}$ holds for an arbitrary scalar field f . Discuss the result.

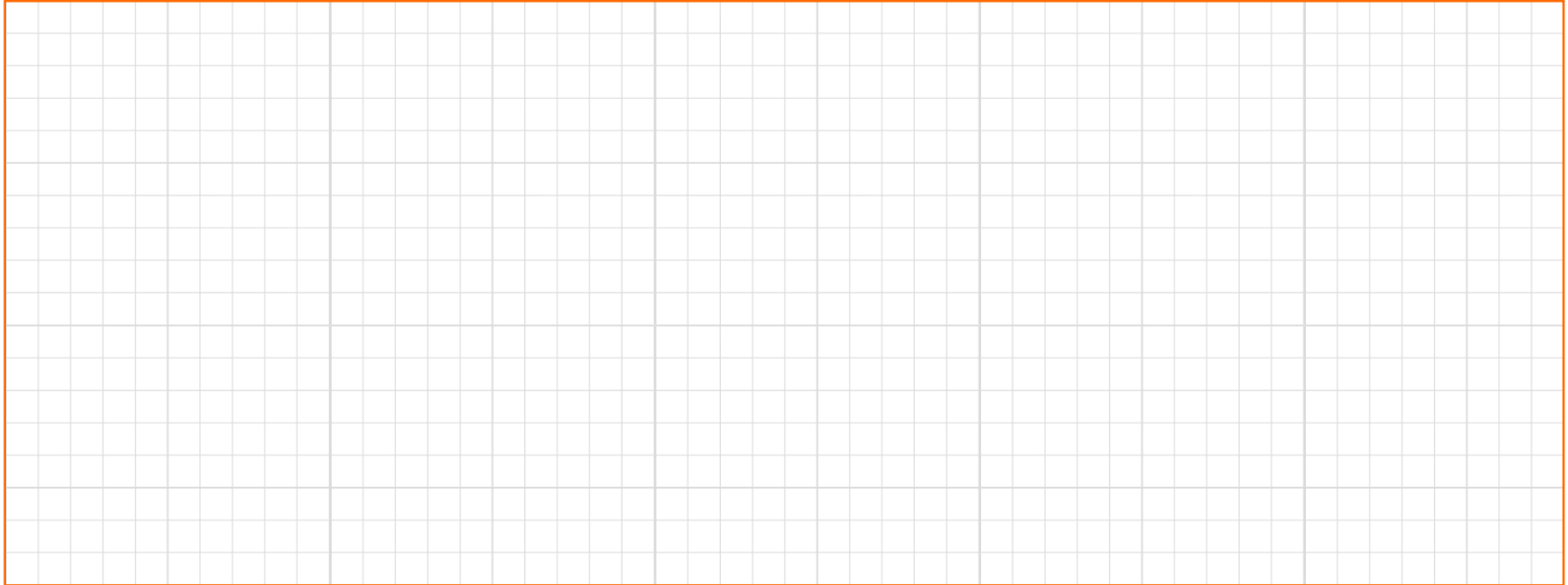


Appendix to 1.2.2: The del operator

Homework : Divergence of rotation is zero

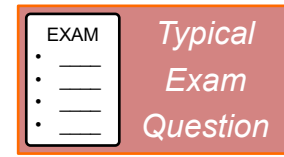


Demonstrate that $\vec{\nabla} (\vec{\nabla} \times \vec{F}) = 0$ holds for an arbitrary vector field. Discuss the result.

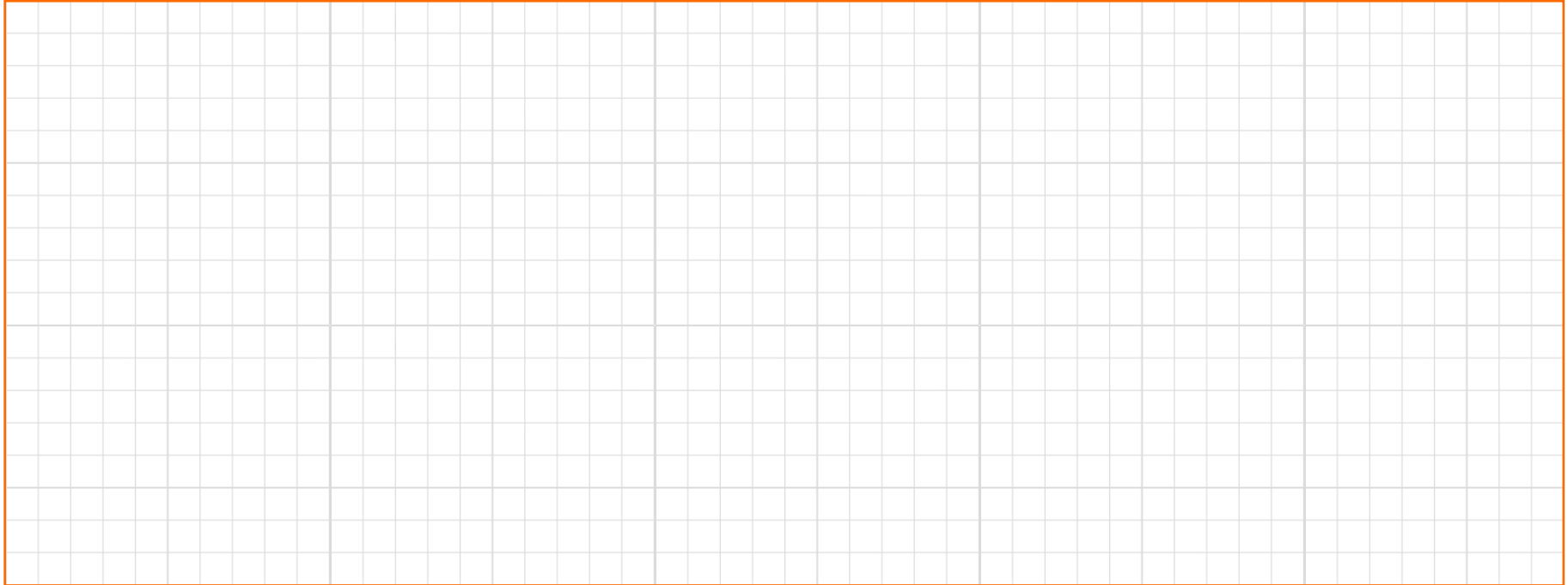


Appendix to 1.2.2: The del operator

Homework: Source-free, rotation-free vector fields

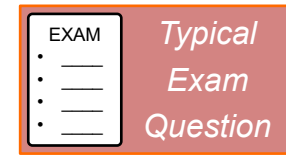


Show that the following vector field $\vec{F} = (y, x, 0)$ is both source-free and rotation-free.

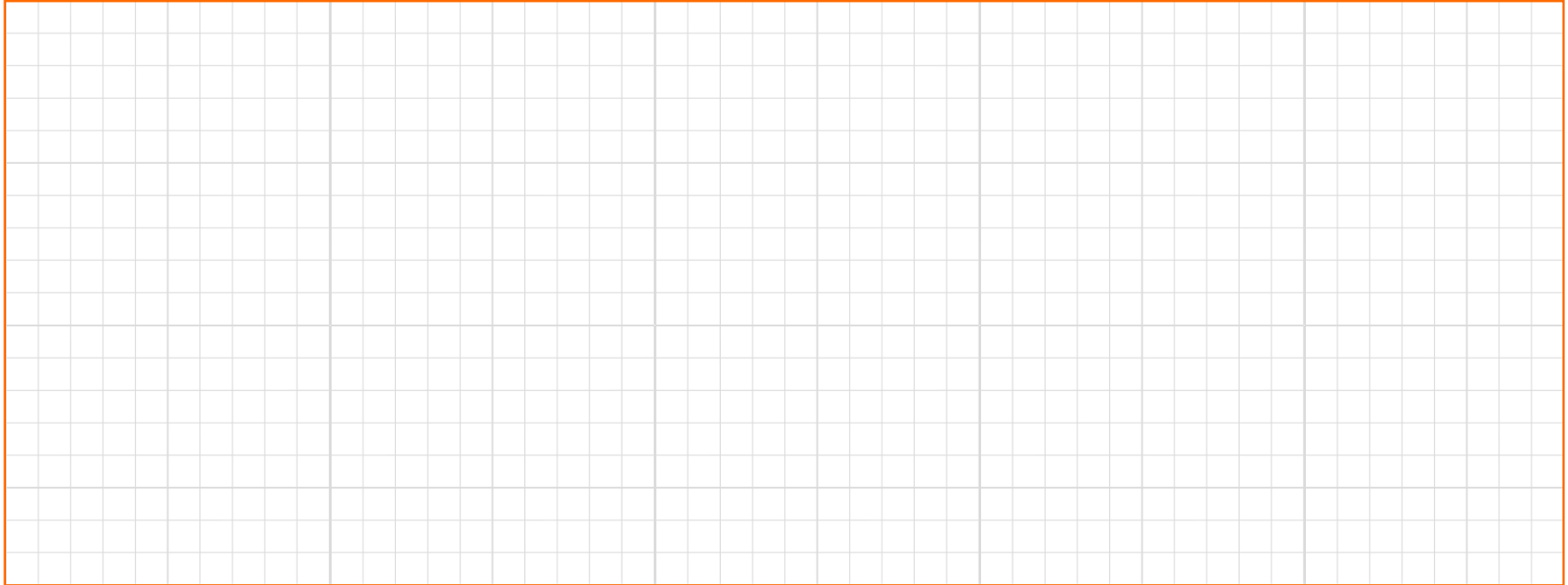


Appendix to 1.2.2: The del operator

Homework: Vector fields with both divergence and rotation



Show that the following vector field $\vec{F} = (-y, x^2, z)$ has both divergence and rotation. Calculate divergence of the rotation.



Summary

- Scalar field: $f: \mathbb{R}^n \rightarrow \mathbb{R}$.
- Vector field: $\vec{F}: \mathbb{R}^n \rightarrow \mathbb{R}^n$.
- $\text{grad } f = \vec{\nabla} f$ is a vector field.
- $\text{div } \vec{F} = \vec{\nabla} \cdot \vec{F}$ is a scalar field (net outflow density).
- $\text{curl } \vec{F} = \vec{\nabla} \times \vec{F}$ is a vector field (local rotation/circulation density).
- Identities: $\vec{\nabla} \times (\vec{\nabla} f) = \vec{0}$ and $\vec{\nabla} \cdot (\vec{\nabla} \times \vec{F}) = 0$.
- A field can be both divergence-free and rotation-free (e. g. $\vec{F} = (y, x, 0)$).
- A field can have both divergence and rotation (e. g. $\vec{F} = (-y, x^2, z)$).

1.2 Vector calculus

Version

Version 4.0.0

- Initial version summer semester 2026

4.0.1

- Error correction p. 9

4.0.2

- Consistent symbols for surface and boundary

4.1.0

- Exercise on grad in spherical coordinates and equation of rot updated